

A sharp Fourier certificate for planar circle packing

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Abstract

We resolve the planar Cohn–Elkies sharpness conjecture: the two-point Fourier bound attains the optimal circle-packing density $\pi/(2\sqrt{3})$. We construct a radial Schwartz certificate and prove its global sign conditions using rigorous interval arithmetic and analytic estimates. The same certificate recovers the classical uniqueness of the triangular packing among periodic equality cases.

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1 Introduction

The planar circle-packing problem asks what fraction of the plane can be occupied by congruent disks with disjoint interiors. Scale their radius to $1/2$, so their centers form a set $\mathcal{C} \subset \mathbb{R}^2$ with $|x - y| \geq 1$ for distinct centers. If B_R is the disk of radius R centered at the origin, we use the upper covered-area density

$$\bar{\delta}(\mathcal{C}) = \limsup_{R \rightarrow \infty} \frac{\text{vol}(B_R \cap \bigcup_{c \in \mathcal{C}} B(c, 1/2))}{\text{vol } B_R}.$$

The triangular lattice attains $\pi/(2\sqrt{3})$, and Thue's theorem gives the matching upper bound. Hales presents an elementary geometric proof based on an idea of Rogers [5, pp. 442–443].

The Fourier-analytic approach of Cohn and Elkies [2] asks whether the same optimum can be certified by one function with compatible signs before and after Fourier transformation. The density theorem alone does not supply such a function. Our result is a sharp Fourier certificate for the classical optimum.

Throughout, the Fourier transform is

$$\hat{f}(\xi) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i x \cdot \xi} \, dx.$$

For real radial integrable functions it is real and radial. A Schwartz function is a smooth function whose derivatives decay faster than every inverse power of the radius; this regularity is stronger than the admissibility conditions in the Cohn–Elkies theorem.

Theorem 1.1 (Sharp Fourier certificate). *There is a real radial Schwartz function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that*

$$\hat{f}(0) = 1, \quad f(0) = \frac{2}{\sqrt{3}}, \quad \hat{f}(\xi) \geq 0 \quad (\xi \in \mathbb{R}^2), \quad f(x) \leq 0 \quad (|x| \geq 1). \quad (1.1)$$

Theorem 1.1 gives an affirmative answer to the two-dimensional case of Cohn and Elkies's Conjecture 7.3 [2], concerning sharp auxiliary functions for their linear programming bound. Their Definition 2.2 and Theorem 3.1 apply to this Schwartz function and to arbitrary packings, not only periodic ones. Their opposite Fourier phase gives the same transform for these even functions. For disks of radius $1/2$ they give

$$\text{vol } B(0, 1/2) \frac{f(0)}{\hat{f}(0)} = \frac{\pi}{4} \frac{2}{\sqrt{3}} = \frac{\pi}{2\sqrt{3}}.$$

This is the density of the lattice generated by $(1, 0)$ and $(1/2, \sqrt{3}/2)$. Thus the theorem identifies an extremizer for this analytic bound. To put it in the equal-origin formulation of Cohn and Elkies's Theorem 3.2, which their Conjecture 7.3 invokes, set

$$g(x) = \frac{\sqrt{3}}{2} f((\sqrt{3}/2)^{1/2} x).$$

The two-dimensional scaling rule gives $g(0) = \hat{g}(0) = 1$, $\hat{g} \geq 0$, and $g(x) \leq 0$ for $|x| \geq (2/\sqrt{3})^{1/2}$, exactly the requested planar witness. The sign estimates also determine the zeros in their required domains. For f , every zero with $|x| \geq 1$ has integral squared radius. This restriction recovers a classical equality consequence: among periodic packings, equality in the density bound forces the triangular lattice (Corollary 5.2). The equal-origin witness g has zeros outside the shells of the triangular lattice of covolume one, so it does not meet the additional zero-set requirement of Cohn and Elkies's Conjecture 8.1. Section 2 gives an explicit extra node, and Section 5 records the zero sets in their respective sign domains and final radial coordinates.

1.1 Fourier certificates and interpolation

Cohn and Elkies formulated their sharpness conjecture in dimensions 2, 8, and 24 [2, Conjecture 7.3]. Their finite Laguerre–Gaussian approximations already combine prescribed double roots, numerical searches for the remaining parameters, and rigorous sign checks by rational arithmetic and Sturm root counts [2, Section 7, pp. 704–705]. Exact attainment of the conjectured sharp bound remained unresolved in that approach. Viazovska constructed the required function in dimension 8 [10, Theorems 3 and 4], and Cohn, Kumar, Miller, Radchenko, and Viazovska constructed its dimension-24 counterpart [3, Section 4]. Their constructions use modular forms to control a function and its Fourier transform simultaneously. Cohn and Elkies report that Gorbachev independently obtained essentially the same linear programming bound [2, p. 694].

Interpolation gives a broader setting for these constructions. Radchenko and Viazovska recovered even Schwartz functions on the real line from their values and Fourier values at the square roots of nonnegative integers [7, Theorem 1]. Cohn, Kumar, Miller, Radchenko, and Viazovska developed interpolation using values and first derivatives in dimensions 8 and 24 and used it to prove universal optimality of the E_8 and Leech lattices [4, Theorems 1.4, 1.7, and 1.9].

The planar shell data have a different feature. Cohn, Kumar, Miller, Radchenko, and Viazovska conjectured that values and first derivatives of a radial Schwartz function and its Fourier transform on the nonzero shells of the triangular lattice normalized to covolume one do not determine the function uniquely [4, Conjecture 7.5]. Sardari proved this nonuniqueness by constructing infinitely many linearly independent radial Schwartz functions with double zeros, before and after Fourier transformation, on those shells [8, Theorem 1.7]. Thus prescribing the shell data alone does not single out a planar certificate. We solve a specified interpolation problem within an explicit family of Gaussian-damped cardinal functions and establish the signs separately.

A 2026 bachelor’s thesis abstract by Zhitniiaia [12] announces an integral representation of a magic function for the hexagonal lattice using weakly holomorphic modular forms and theta functions for $\Gamma_0(12)$. Our comparison is limited to the institutional abstract, which does not specify the complete sign, normalization, and regularity statement. The precise relation of that claim to Theorem 1.1 remains to be determined from the full thesis.

1.2 The construction

For a sharp Fourier bound on a lattice, Poisson summation forces the auxiliary function and its Fourier transform to vanish on the nonzero shells of the lattice and its dual, respectively [2, Section 5]. After normalizing the triangular lattice to covolume one, it is isometric to its dual, so the two sets of shell radii coincide. This normalization makes the coupled Fourier equations symmetric. At the end we dilate the resulting function to put the first physical-space shell at the exclusion radius one in Theorem 1.1. We cover their squared radii, in a fixed scaled coordinate, by a periodic set of interpolation nodes. Periodic supersets of the values of the triangular quadratic form also occur in Sardari’s construction [8, Section 6.1]; here a fixed periodic set permits an explicit trigonometric cardinal family.

An interior zero of a differentiable function with a fixed sign must also have zero first derivative. We therefore impose zeros of order at least two on the Fourier side and at every physical-space shell beyond the first. The first physical-space shell is the boundary of the required sign region, so a simple zero is possible there; its prescribed negative derivative will fix the normalization.

A trigonometric polynomial vanishes doubly on the chosen set. Dividing it by linear and quadratic factors produces functions that prescribe values and first derivatives at individual nodes. This Hermite-cardinal pattern has a precedent in the Gaussian Beurling–Selberg constructions of

Carneiro, Littmann, and Vaaler [1, formulas (1.2)–(1.3)]; the Fourier coupling and sign estimates below are proved for the present family. Multiplication by a fixed Gaussian gives radial Schwartz functions. Their Fourier transforms admit integral representations with additional Gaussian decay. We use that decay to solve the coupled interpolation equations: a finite matrix handles the first nodes, and explicit estimates control the remaining infinite system. The inversion acts on the full unweighted ℓ^1 coefficient space. Its uniqueness concerns the prescribed coefficient lists within this chosen family.

The resulting functions have exact zeros, but the coefficient lists used for computation are finite approximations. An explicit per-list ℓ^1 error bound connects these approximations to the exact solution. Near a zero, a raw error divided by its vanishing power would be singular. We instead subtract the lower Taylor terms of the error before division; positive Bernstein coefficients then certify the signs on complete half-gaps. A separate sine-product barrier and second-derivative estimate control the entire unbounded region. Poisson summation first equates the two origin values. Differentiating its dilation identity then determines their common positive value from the prescribed first-shell derivative, so the final normalization is exact.

Section 2 develops the cardinal functions and their Fourier transforms. Section 3 solves the interpolation equations, and Section 4 proves the global signs and identifies their zeros. Section 5 establishes the exact normalization, performs the final dilation, and derives the packing consequences. Appendix A gives the finite data and the primary interval certificate, including the analytic integration error that connects its finite sums to exact moments. Appendix B gathers the rational alternatives and independent checks, with their precise mathematical and computational scopes.

2 Gaussian cardinal functions

The construction uses a family of entire functions whose values and first derivatives can be prescribed at a periodic set of nodes. Multiplication by a fixed Gaussian turns these functions into radial Schwartz functions. Their Fourier transforms have strictly stronger Gaussian decay, which will make the coupled interpolation equations tractable.

Set

$$b = \frac{\sqrt{3}}{2}, \quad h = \frac{2}{5}, \quad B = b^{-2} = \frac{4}{3}, \quad s = b|x|^2. \quad (2.1)$$

The same coordinate, $s = b|\xi|^2$, is used on the Fourier side. To see the lattice geometry behind this choice, let Λ have basis

$$u = \frac{1}{\sqrt{b}}(1, 0), \quad v = \frac{1}{\sqrt{b}}(1/2, b). \quad (2.2)$$

This oriented basis has determinant one. If $\mathcal{R}$ is counterclockwise rotation through 90 degrees, its dual basis is $-\mathcal{R}v, \mathcal{R}u$. Hence $\Lambda^* = \mathcal{R}\Lambda$: the lattice and its dual have the same shell radii. For $a = ju + kv$, with $j, k \in \mathbb{Z}$, the squared radius in the common coordinate is

$$s_a := b|a|^2 = j^2 + jk + k^2. \quad (2.3)$$

We impose interpolation conditions at the positive nodes of a periodic set:

$$A = \{0, 1, 3, 4, 7, 9\}, \quad \mathcal{N} = (12\mathbb{Z} + A) \cap (0, \infty). \quad (2.4)$$

The set $\mathcal{N}$ contains every positive value of $j^2 + jk + k^2$. Indeed, $j^2 + jk + k^2 \equiv (j - k)^2 \pmod{3}$, so it is never $2 \pmod{3}$. Modulo 4, it is 0 when j, k are both even and is odd otherwise. Combining these restrictions gives precisely the six allowed residues in A . The inclusion is strict: $15 \in \mathcal{N}$, but

$j^2 + jk + k^2 = 15$ would imply $(2j + k)^2 \equiv 2k^2 \pmod{5}$. Since 2 is not a square modulo 5, this forces $5 \mid k$ and then $5 \mid j$, which would make 25 divide 15. Thus we require zeros at some nodes that are not lattice shell coordinates. This deliberate enlargement uses a periodic residue pattern with an elementary trigonometric vanishing function. Covering planar shell values by periodic sets also occurs in Sardari's Fourier interpolation construction [8, Section 6.1]; the Gaussian transformation and the coupled inversion needed here will be proved directly.

2.1 The cardinal family

The values and first derivatives will be encoded by double-pole and simple-pole terms multiplied by a common vanishing factor. This is the same cardinal pattern that occurs in Gaussian Beurling–Selberg interpolation [1, formulas (1.2)–(1.3)]. The nodes and simultaneous Fourier constraints here are different.

For $-6 \leq j \leq 6$, put $t_j = j/6$, and define

$$P(s) = \prod_{a \in A} \left(2 \sin \frac{\pi(s-a)}{12} \right)^2 = \sum_{j=-6}^6 P_j e^{i\pi t_j s}. \quad (2.5)$$

Here $P_{-j} = \overline{P_j}$, and

$$(P_0, P_1, \dots, P_6) = \left(5, -1 + 2ib, 1 + 2ib, -2, -\frac{1}{2} + ib, -1 - 2ib, 1 \right). \quad (2.6)$$

For example, with $w = e^{i\pi s/6}$ and $\rho = e^{i\pi/6}$, each factor is

$$\left(2 \sin \frac{\pi(s-a)}{12} \right)^2 = -\rho^{-a} w^{-1} (w - \rho^a)^2.$$

The six minus signs cancel, and $\rho^{-\sum_{a \in A} a} = \rho^{-24} = 1$. Thus the product in (2.5) equals

$$w^{-6} \prod_{a \in A} (w - \rho^a)^2.$$

Expanding this Laurent polynomial gives (2.6).

Every zero $n \in 12\mathbb{Z} + A$ of P is double. Define Q_n, D_n by

$$P(n+u) = Q_n u^2 (1 + D_n u + O(u^2)). \quad (2.7)$$

If $a \in A$ is the residue of n modulo 12, then

$$\begin{aligned} Q_n &= \left(\frac{\pi}{6} \right)^2 \prod_{\substack{a' \in A \\ a' \neq a}} \left(2 \sin \frac{\pi(a-a')}{12} \right)^2, \\ D_n &= \frac{\pi}{6} \sum_{\substack{a' \in A \\ a' \neq a}} \cot \frac{\pi(a-a')}{12}. \end{aligned} \quad (2.8)$$

Thus Q_n, D_n are periodic in n , and their exact values are

a	Q_n	D_n
0	$\pi^2/3$	$-7\pi\sqrt{3}/18$
1	$(2\pi^2/3)(2 - \sqrt{3})$	$\pi(3 + \sqrt{3})/18$
3	$(2\pi^2/3)(2 - \sqrt{3})$	$-\pi(3 + \sqrt{3})/18$
4	$\pi^2/3$	$7\pi\sqrt{3}/18$
7	$(2\pi^2/3)(2 + \sqrt{3})$	$-\pi(3 - \sqrt{3})/18$
9	$(2\pi^2/3)(2 + \sqrt{3})$	$\pi(3 - \sqrt{3})/18$

The expansion follows by isolating the vanishing sine factor; logarithmic differentiation of the remaining factors gives D_n . The table then uses the elementary trigonometric values at multiples of $\pi/12$. In particular,

$$Q_n > \frac{7}{4} = 1.75, \quad |D_n| < \frac{53}{25} = 2.12. \quad (2.9)$$

To check these inequalities with rational bounds, use $157/50 < \pi < 22/7$ and $\sqrt{3} < 26/15$. The minimum of Q_n is greater than $197192/112500 > 7/4$, and the maximum of $|D_n|$ is less than $286/135 < 53/25$.

Let $C \in \mathbb{R}$, and let $(c_n, d_n)_{n \in \{0\} \cup \mathcal{N}}$ be a real coefficient list with

$$\|(c, d)\|_1 := \sum_{n \in \{0\} \cup \mathcal{N}} (|c_n| + |d_n|) < \infty. \quad (2.10)$$

Away from the nodes, define

$$p(s) = P(s)R(s), \quad R(s) = C + \sum_{n \in \{0\} \cup \mathcal{N}} \left(\frac{c_n}{(s-n)^2} + \frac{d_n}{s-n} \right). \quad (2.11)$$

We next show that the apparent singularities are removable, with uniform control of the resulting entire function.

Lemma 2.1 (Finite-measure representation). *The function in (2.11) extends to an entire function with the representation*

$$p(s) = \int_{-1}^1 e^{i\pi ts} d\mu(t), \quad s \in \mathbb{C}, \quad (2.12)$$

where μ is a finite complex measure, supported in $[-1, 1]$, with $\mu(-E) = \overline{\mu(E)}$ for Borel sets E . Its total variation satisfies

$$\|\mu\|_{\text{TV}} \leq 25|C| + \frac{65\pi^2}{18} \sum_n |c_n| + \frac{32\pi}{3} \sum_n |d_n|. \quad (2.13)$$

The atoms of μ have mass CP_j at t_j , for $-6 \leq j \leq 6$. For real-axis evaluations, the remaining measure can be folded onto $[0, 1]$: on $t_{j-1} < t < t_j$, its folded density is $\sum_n (c_n u_n(t) + d_n v_n(t))$, where

$$\begin{aligned} u_n(t) &= -2\pi^2 e^{-i\pi tn} \sum_{l=j}^6 (t_l - t) P_l e^{i\pi t_l n}, \\ v_n(t) &= 2i\pi e^{-i\pi tn} \sum_{l=j}^6 P_l e^{i\pi t_l n}. \end{aligned} \quad (2.14)$$

The folded atoms have masses CP_0 at 0 and $2CP_j$ at t_j , $1 \leq j \leq 6$. Integrate against this folded measure and take the real part to obtain $p(s)$ for real s , or any of its real-axis derivatives.

Proof. Fix a node n and write $u = s - n$. For real τ , the identities

$$\begin{aligned} \frac{e^{i\pi\tau u} - 1}{u} &= i\pi \int_0^\tau e^{i\pi t u} dt, \\ \frac{e^{i\pi\tau u} - 1 - i\pi\tau u}{u^2} &= -\pi^2 \int_0^\tau (\tau - t) e^{i\pi t u} dt \end{aligned} \quad (2.15)$$

hold for complex u , with the values at zero taken by continuity. When $\tau < 0$, the integrals are oriented integrals. Inserting $\tau = t_l$ and multiplying by $P_l e^{i\pi t_l n}$, the subtracted terms sum to zero because $P(n) = P'(n) = 0$. On the positive interval (t_{j-1}, t_j) , only frequencies with $l \geq j$ contribute. The resulting density for $P(s)/(s - n)^2$ is $u_n/2$, and that for $P(s)/(s - n)$ is $v_n/2$. The negative-frequency densities are their conjugate reflections, which proves (2.14) and accounts for its factors of two.

All phase factors involving n have modulus one on the real integration interval. Hence the measures for the two individual basis functions satisfy

$$\begin{aligned} \int_0^1 |u_n(t)| dt &\leq 2\pi^2 \sum_{l=1}^6 |P_l| \int_0^{t_l} (t_l - t) dt = \pi^2 \sum_{l=1}^6 |P_l| t_l^2 = \frac{65\pi^2}{18}, \\ \int_0^1 |v_n(t)| dt &\leq 2\pi \sum_{l=1}^6 |P_l| t_l = \frac{32\pi}{3}. \end{aligned}$$

These are also bounds for the total variations of the full measures on $[-1, 1]$. The atomic measure for CP has total variation $|C| \sum_{j=-6}^6 |P_j| = 25|C|$. The coefficient summability therefore gives convergence of the sum of basis measures in total variation and proves (2.13).

For any compact subset of $\mathbb{C}$, the integrand in (2.12), and each of its derivatives with respect to s , is bounded uniformly in $t \in [-1, 1]$. The integral is consequently entire, and the finite cardinal sums converge to it locally uniformly. Away from the nodes, the series in (2.11) converges locally uniformly and has the same limit. Thus the integral supplies its entire extension. Conjugate symmetry of the measure makes that extension real on the real axis. $\square$

In particular, for every integer $k \geq 0$,

$$\left| p^{(k)}(s) \right| \leq \pi^k \|\mu\|_{TV} \quad (s \in \mathbb{R}). \quad (2.16)$$

The finite cardinal sums and their derivatives converge locally uniformly: the assertion for derivatives follows either directly from the measure bound or from Cauchy's integral formula on a slightly larger compact set. Passing their node values and derivatives to the limit in (2.7) gives the cardinal identities

$$p(n) = Q_n c_n, \quad p'(n) = Q_n (d_n + D_n c_n) \quad (n \in \{0\} \cup \mathcal{N}). \quad (2.17)$$

The other summands in (2.11) vanish to second order at n , which explains why these identities involve only its own pair of coefficients.

2.2 Fourier transformation

For a cardinal function p , let

$$q_p(x) = e^{-\pi h s} p(s), \quad s = b|x|^2.$$

Define, for real $-1 \leq t \leq 1$,

$$\lambda(t) = \frac{i}{b(t+ih)}, \quad z(t) = -\frac{B}{t+ih} - ih, \quad (2.18)$$

and, using the measure from Lemma 2.1, put

$$K_p(s) = \int_{-1}^1 \lambda(t) e^{i\pi z(t)s} d\mu(t). \quad (2.19)$$

Proposition 2.2 (Gaussian Fourier pairing). *The function q_p is real, radial, and Schwartz on $\mathbb{R}^2$, and*

$$\widehat{q_p}(\xi) = e^{-\pi h s} K_p(s), \quad s = b|\xi|^2.$$

The function K_p is entire and real on the real axis. Moreover,

$$\operatorname{Im} z(t) = h \left(\frac{B}{t^2 + h^2} - 1 \right) \geq \frac{26}{435} > 0 \quad (-1 \leq t \leq 1). \quad (2.20)$$

Consequently every derivative of $K_p(s)$ decays exponentially as $s \rightarrow +\infty$.

Proof. By (2.12),

$$q_p(x) = \int_{-1}^1 e^{-\pi b(h-it)|x|^2} d\mu(t).$$

Every spatial derivative of the integrand is a polynomial in x , whose coefficients are bounded uniformly for $t \in [-1, 1]$, times a Gaussian of modulus $e^{-\pi b h |x|^2}$. The finite variation of μ justifies differentiation under the integral and bounds every Schwartz seminorm. Reality follows from the real values of p on the real axis, and radially from the dependence on $|x|^2$.

For $\operatorname{Re} a > 0$, the two-dimensional Gaussian transform in our Fourier convention is

$$\widehat{e^{-\pi a |x|^2}}(\xi) = a^{-1} e^{-\pi |\xi|^2 / a}.$$

This follows from the one-dimensional Gaussian integral by taking products and analytically continuing in a to the right half-plane. Apply this with $a = b(h - it)$. Fubini's theorem applies because the integrands have a common integrable Gaussian majorant. We have

$$a^{-1} = \lambda(t), \quad -\frac{\pi |\xi|^2}{a} + \pi h s = i\pi z(t)s,$$

which gives (2.19) and the asserted Fourier transform. The curve $z([-1, 1])$ is compact, so K_p is entire by differentiation under its defining integral. Also,

$$\lambda(-t) = \overline{\lambda(t)}, \quad z(-t) = -\overline{z(t)}.$$

Thus the same folded measure and real-part convention as in Lemma 2.1 apply to K_p and its real-axis derivatives.

Finally, the imaginary part in (2.20) is smallest at $|t| = 1$, where its value is $26/435$. Differentiating (2.19) gives, for $s \geq 0$,

$$\left| K_p^{(k)}(s) \right| \leq \|\mu\|_{\text{TV}} \max_{|t| \leq 1} (|\lambda(t)| (\pi |z(t)|)^k) e^{-26\pi s/435}.$$

This proves the decay assertion. □

No cardinal representation of K_p is assumed. Only its entireness, its values and derivatives, and the estimates obtained from (2.19) will be used in the interpolation equations. In particular, the measure in that formula is supported on the real interval $[-1, 1]$; later complex-disk estimates concern specified finite integrands, not an analytic continuation in the integration variable of an arbitrary ℓ^1 density.

Remark 2.3 (A real-radius Hankel representation). For real $s \geq 0$, the same transform has the representation

$$K_p(s) = e^{\pi h s} \frac{\pi}{b} \int_0^\infty e^{-\pi h u} p(u) J_0\left(\frac{2\pi\sqrt{us}}{b}\right) du, \quad (2.21)$$

where $J_0(v) = (2\pi)^{-1} \int_0^{2\pi} e^{-iv \cos \theta} d\theta$ for real v . To prove this, use polar coordinates in the Fourier integral for q_p , perform the angular integral, and put $u = br^2$. The resulting radial integral is $\widehat{q}_p(\xi) = (\pi/b) \int_0^\infty e^{-\pi h u} p(u) J_0(2\pi\sqrt{us}/b) du$, with $s = b|\xi|^2$. Absolute convergence follows from $|p(u)| \leq \|\mu\|_{TV}$ and $|J_0(v)| \leq 1$. Multiplication by $e^{\pi h s}$ gives (2.21). This is an identity on the nonnegative real axis; it is not used as a numerical sign certificate below.

2.3 The coupled interpolation conditions

We use two cardinal functions p_1, p_2 , with coefficients $c_{i,n}, d_{i,n}, C_i$ for function index $i \in \{1, 2\}$. Their coefficients at positive nodes will be determined later; fix

$$(c_{1,0}, d_{1,0}, C_1) = (1, 0.44, -0.013), \quad (c_{2,0}, d_{2,0}, C_2) = (0, -0.368, 0.017). \quad (2.22)$$

All terminating decimals in the construction denote exact rational numbers. Write $K_i = K_{p_i}$ and $H_i = p_i + K_{3-i}$. The associated pair is

$$F(x) = e^{-\pi h s} H_1(s), \quad \widehat{F}(\xi) = e^{-\pi h s} H_2(s), \quad s = b|x|^2 \text{ or } b|\xi|^2, \quad (2.23)$$

respectively. Indeed, $F = q_{p_1} + \widehat{q_{p_2}}$, and Fourier inversion gives $\widehat{F} = \widehat{q_{p_1}} + q_{p_2}$, since the functions are even. Thus both functions in (2.23) are real, radial, and Schwartz for every pair of admissible coefficient lists.

The required interpolation conditions are

$$\begin{aligned} H_1(n) &= H_2(n) = 0, \\ H'_2(n) &= 0, \quad H'_1(n) = -Q_n \mathbf{1}_{\{n=1\}} \quad (n \in \mathcal{N}). \end{aligned} \quad (2.24)$$

Here primes denote derivatives with respect to s . They give zeros of order at least two for both functions at every positive node except that H_1 has a prescribed simple zero at 1.

Lemma 2.4 (Coefficient form of the interpolation). *For two real ℓ^1 coefficient lists, conditions (2.24) are equivalent to*

$$\begin{aligned} c_{i,n} &= -\frac{K_{3-i}(n)}{Q_n}, \\ d_{i,n} &= \frac{D_n K_{3-i}(n) - K'_{3-i}(n)}{Q_n} - \mathbf{1}_{\{i=1, n=1\}} \quad (i = 1, 2, n \in \mathcal{N}). \end{aligned} \quad (2.25)$$

Proof. By (2.17),

$$H_i(n) = Q_n c_{i,n} + K_{3-i}(n), \quad H'_i(n) = Q_n (d_{i,n} + D_n c_{i,n}) + K'_{3-i}(n).$$

The first equation in (2.25) is therefore equivalent to $H_i(n) = 0$. Substituting it into the second displayed identity gives the derivative condition in (2.24) exactly when the second coefficient equation holds. Division by Q_n is legitimate by (2.9). $\square$

3 Solving the interpolation equations

The equations (2.25) couple two coefficient lists through the transformed part of the construction. We solve them on ℓ^1 , using the strict damping of the Fourier kernel to control every row beyond a fixed finite set.

Let

$$X = \ell^1(\mathcal{N}; \mathbb{R}^2), \quad \|x\|_1 = \sum_{n \in \mathcal{N}} (|c_n| + |d_n|) \quad \text{for } x = ((c_n, d_n))_{n \in \mathcal{N}}.$$

When writing matrices, we order all c coordinates first and all d coordinates second, with increasing nodes within each group. All operator norms below are the induced ℓ^1 norms, equivalently the maximum absolute column sum for a finite matrix. A subscript specifying a set of nodes includes both coordinates at each node; a star denotes all positive nodes.

For $x \in X$, let p_x be the function (2.11) with these positive-node coefficients and with $C = c_0 = d_0 = 0$. Define the real linear operator S by

$$(Sx)_{c,n} = -\frac{K_{p_x}(n)}{Q_n}, \quad (Sx)_{d,n} = \frac{D_n K_{p_x}(n) - K'_{p_x}(n)}{Q_n}. \quad (3.1)$$

The bounds proved below show, in particular, that S is a bounded operator on X .

We use two successive decompositions of this operator. Set

$$J = \{1, 3, 4\}, \quad E = \mathcal{N} \cap [7, 100], \quad I_f = J \cup E = \mathcal{N} \cap [1, 100], \quad D = S_{J,J}. \quad (3.2)$$

A finite inverse handles the coordinates on J . A first Schur complement then incorporates E , using separate bounds for its rows through 16 and beyond 16; a second incorporates the infinite tail beyond 100. The tail bounds apply to output rows uniformly over all input columns. The first output rows require a separate, larger bound even for columns indexed by large nodes.

3.1 Uniform envelopes for the Fourier terms

We use the following rational upper and lower bounds. As throughout the construction, every finite decimal denotes its exact rational value.

j	b_j^*	a_j^*	g_j	M_j	β_j
0	2.887	2.934	2.933	—	—
1	2.665	2.713	2.440	56	0
2	2.218	2.269	1.567	48	4.8
3	1.804	1.859	.900	35	3.17
4	1.486	1.548	.482	25	1.946
5	1.250	1.320	.224	19	1.217
6	1.073	1.151	.0597	6.284	.792

Table 1: Envelopes for the folded Fourier integrals.

At each endpoint $t_j = j/6$,

$$|\lambda(t_j)| \leq b_j^*, \quad |z(t_j)| \leq a_j^*, \quad \text{Im } z(t_j) \geq g_j.$$

On the interval $t_{j-1} \leq t \leq t_j$, the corresponding bounds are

$$|\lambda(t)| \leq b_{j-1}^*, \quad |z(t)| \leq a_{j-1}^*, \quad \text{Im } z(t) \geq g_j + \beta_j(t_j - t), \quad |u_n(t)|, |v_n(t)| \leq M_j. \quad (3.3)$$

The density bounds hold uniformly for $n \in \{0\} \cup \mathcal{N}$.

Here are direct ways to verify these estimates. The identities

$$|\lambda(t)| = \frac{1}{b\sqrt{t^2 + h^2}}, \quad |z(t)|^2 = h^2 + \frac{B^2 - 2Bh^2}{t^2 + h^2}, \quad \text{Im } z(t) = h \left(\frac{B}{t^2 + h^2} - 1 \right)$$

show that the first two quantities decrease on $[0, 1]$. The nonnegative quantity $2hBt/(t^2 + h^2)^2$, which is minus the derivative of $\text{Im } z$, has its minimum on each of the six intervals at an endpoint. Its only interior critical point on the positive axis is a maximum. These observations give the first three bounds in (3.3) by endpoint substitution. For the densities, $e^{i\pi t_l n}$ depends only on n modulo 12, whereas $|e^{-i\pi t n}| = 1$ on the real interval. The norm of an affine function of t is convex. Thus an upper bound for both folded densities on the j th interval is

$$2 \max_{a \in A} \max \left\{ \pi \left| \sum_{l=j}^6 P_l e^{i\pi t_l a} \right|, \quad \pi^2 \max_{u \in \{t_{j-1}, t_j\}} \left| \sum_{l=j}^6 (t_l - u) P_l e^{i\pi t_l a} \right| \right\}. \quad (3.4)$$

Substitution gives a value less than the displayed M_j in every case. For the constant input $C = 1$, the absolute folded atom masses are

$$(m_0^*, \dots, m_6^*) = (5, 4, 4, 4, 2, 4, 2).$$

For $s \geq 0$, set

$$\begin{aligned} L_j(s) &= b_{j-1}^* e^{-\pi g_j s} \min \left\{ \frac{1}{6}, \frac{1}{\pi \beta_j s} \right\} \quad (1 \leq j \leq 6), \\ L_j^P(s) &= m_j^* b_j^* e^{-\pi g_j s} \quad (0 \leq j \leq 6). \end{aligned} \quad (3.5)$$

When $\beta_j s = 0$, the minimum in the first line is understood to be $1/6$. Integrating the exponential envelope in (3.3) over a piece of length $1/6$ gives its factor $L_j(s)$. Differentiating the kernel contributes $(i\pi z(t))^\ell$. Consequently, for every integer $\ell \geq 0$ and every input p of the form (2.11),

$$\left| K_p^{(\ell)}(s) \right| \leq w \sum_{j=1}^6 M_j L_j(s) (\pi a_{j-1}^*)^\ell + |C| \sum_{j=0}^6 L_j^P(s) (\pi a_j^*)^\ell, \quad s \geq 0, \quad (3.6)$$

where $w = \sum_{n \in \{0\} \cup \mathcal{N}} (|c_n| + |d_n|)$. The argument applies first to finite lists and then to all such lists by the finite-measure representation and absolute summability.

Both Q_n and D_n are 12-periodic on the nodes. Moreover,

$$L_j(n + 12k) \leq e^{-12k\pi g_j} L_j(n) \quad (k \geq 0),$$

because the minimum factor in (3.5) is nonincreasing. For every real cutoff $x \geq 0$, the value and derivative terms in (3.1), together with (3.6), therefore imply

$$\|S_{>x,*}\| \leq \sum_{n \in \mathcal{N} \cap (x, x+12]} \sum_{j=1}^6 \frac{1 + |D_n| + \pi a_{j-1}^*}{Q_n} \frac{M_j L_j(n)}{1 - e^{-12\pi g_j}}. \quad (3.7)$$

Here $> x$ means a strict inequality for the row node. This is a bound uniform over all input columns. It applies as well to a single input coefficient c_0 or d_0 .

For completeness, this estimate also justifies the operator definition on the whole space. Apply the bound first to one positive-node basis column, then sum with the absolute coefficients of a

finitely supported list. The case $x = 0$ bounds the resulting output norm by a constant times the input norm. Finite lists are dense in X , so this map has a unique bounded extension to X . The basis measures converge in total variation by Lemma 2.1; hence $K_{p_x}(n)$ and $K'_{p_x}(n)$ converge at each fixed node as the lists are truncated. The extended map therefore has exactly the coordinates in (3.1), not merely the coordinates of an auxiliary matrix limit.

For $x = 0, 4, 16, 100$, respectively, the right side of (3.7) is less than

$$27.7, \quad .272, \quad .0145, \quad 4.7 \cdot 10^{-10}.$$

We shall use the slightly weaker bounds

$$\|S_{>0,*}\| < 28, \quad \|S_{>4,*}\| < .28, \quad \|S_{>16,*}\| < .015, \quad \|S_{>100,*}\| < 5 \cdot 10^{-10}. \quad (3.8)$$

For the input $C = 1$, the analogue of (3.7) replaces $M_j L_j$ by L_j^P , uses a_j^* in place of a_{j-1}^* , and sums over $0 \leq j \leq 6$. For rows greater than 100, it gives the bound

$$\sum_{n \in \mathcal{N}, n > 100} \left(\frac{|K_P(n)|}{Q_n} + \frac{|D_n K_P(n) - K'_P(n)|}{Q_n} \right) < 3.2 \cdot 10^{-8}. \quad (3.9)$$

The scalar substitutions in these envelope estimates use only the explicit constants and the node data. Section A.6 records the finite scalar bounds and the precise coverage of their supplied verifier.

Remark 3.1 (Compactness of the interpolation operator). Let P_M keep the output coordinates with node at most M . Then $P_M S$ has finite rank. The right side of (3.7) tends to zero as $x \rightarrow \infty$: there are only six nodes in any period, the periodic Q_n, D_n are uniformly bounded with Q_n bounded away from zero, and every $L_j(n)$ contains the factor $e^{-\pi g_j n}$ with $g_j > 0$. Thus $\|S - P_M S\| \rightarrow 0$. The operator S is a norm limit of finite-rank operators and is compact on X . This property describes the tail structure; the explicit inverse below is obtained independently from the quantitative block estimates.

3.2 The finite certificates

There are 51 nodes in I_f and hence 102 scalar coordinates. The approximate lists $\tilde{x}_1, \tilde{x}_2$ are given by Table 2, with the indicated factor 10^{-10} , and are zero at positive nodes beyond 100. Append the fixed data (2.22) to obtain $\tilde{p}_i$, and write $\tilde{K}_i = K_{\tilde{p}_i}$.

Lemma 3.2 (Finite certificates). *For each $\eta \in \{1, -1\}$, the matrix $I - \eta D$ is invertible, and*

$$\|(I - \eta D)^{-1}\| < 32, \quad \|(I - \eta D)^{-1} S_{J,E}\| < 3.7, \quad \|S_{E \cap [7,16], J}\| < .09. \quad (3.10)$$

For $i = 1, 2$ and $n \in I_f$, the table satisfies

$$\begin{aligned} \left| \tilde{c}_{i,n} + \frac{\tilde{K}_{3-i}(n)}{Q_n} \right| &< 10^{-9}, \\ \left| \tilde{d}_{i,n} + \mathbf{1}_{\{i=1, n=1\}} + \frac{\tilde{K}'_{3-i}(n) - D_n \tilde{K}_{3-i}(n)}{Q_n} \right| &< 10^{-9}. \end{aligned} \quad (3.11)$$

For each i , the norm of the tabulated list, including node zero, is less than 2.29, and its norm on nodes greater than 40 is less than .00002.

Proof. The primary interval calculation in Appendix A proves the exact bounds (A.13), (A.14), and (A.15), using the quadrature assertion of Lemma A.1 for the analytic moment remainder. The elementary argument in Section A.4 gives the matrix conclusions, and (A.2) gives the list norms. $\square$

3.3 Inversion on the whole coefficient space

We first prove an inverse bound that keeps finite and tail residuals separate.

Lemma 3.3. *For each $\eta \in \{1, -1\}$, $I - \eta S$ has a bounded inverse on X . If $r = r_f + r_t$, with r_f supported on I_f and r_t on $\mathcal{N} \cap (100, \infty)$, then*

$$\|(I - \eta S)^{-1}r\|_1 \leq 91 \|r_f\|_1 + 2540 \|r_t\|_1. \quad (3.12)$$

Proof. Fix η and first restrict to the finite block I_f . Put

$$V = (I - \eta D)^{-1}, \quad L = S_{E,J}, \quad U = S_{J,E}.$$

The first block of tail estimates and the finite certificates give

$$\|L\| \leq \|S_{E \cap [7,16], J}\| + \|S_{E \cap (16,100], J}\| < .09 + .015 < .11.$$

Eliminating the J coordinates from $(I - \eta S_{I_f, I_f})x_f = r_f$ leaves on E the operator

$$I - \eta S_{E,E} - LVU.$$

Its difference from the identity has norm less than

$$\|S_{E,E}\| + \|L\| \|VU\| < .28 + .11 \cdot 3.7 = .687 < 1.$$

It is therefore invertible by a Neumann series. Solving this block and substituting into the J equations yields an inverse V_f for the whole finite compression, with

$$\begin{aligned} \|V_f\| &\leq \max \left\{ 32 + \frac{(1 + 3.7) \cdot .11 \cdot 32}{1 - .687}, \quad \frac{1 + 3.7}{1 - .687} \right\} \\ &< 90. \end{aligned} \quad (3.13)$$

Indeed, the two entries in the maximum bound the output norm arising from a unit input on J and on E , respectively.

Now write $X = X_f \oplus X_t$, where X_t is the infinite tail space. The full equation $(I - \eta S)x = r$ is equivalent, after elimination of x_f , to

$$(I - \eta S_{t,t} - S_{t,f} V_f S_{f,t})x_t = r_t + \eta S_{t,f} V_f r_f. \quad (3.14)$$

Put $\delta = 5 \cdot 10^{-10}$. By (3.8),

$$\|S_{t,t}\| < \delta, \quad \|S_{t,f}\| < \delta, \quad \|S_{f,t}\| < 28.$$

The operator subtracted from the identity in (3.14) thus has norm less than

$$\delta(1 + 90 \cdot 28) = 2521\delta < 1.$$

Its Neumann series converges in the bounded operators on X_t . Together with the finite inverse this proves invertibility on the full space X , without any passage to a limit of finite-section solutions. Quantitatively,

$$\|x_t\|_1 \leq \frac{\|r_t\|_1 + 90\delta \|r_f\|_1}{1 - 2521\delta}, \quad \|x_f\|_1 \leq 90 \|r_f\|_1 + 2520 \|x_t\|_1.$$

Adding these inequalities gives coefficients less than 91 and 2540 for $\|r_f\|_1$ and $\|r_t\|_1$, respectively, which proves (3.12). $\square$

Proposition 3.4 (Exact interpolation and coefficient enclosure). *There are unique real lists $x_1, x_2 \in X$ satisfying (2.25) with the fixed data (2.22). For each $i = 1, 2$,*

$$\sum_{n \in \mathcal{N}} (|c_{i,n} - \tilde{c}_{i,n}| + |d_{i,n} - \tilde{d}_{i,n}|) < .00003. \quad (3.15)$$

The norm of each exact list, including its node-zero coefficients, is less than 2.3. The corresponding functions satisfy all the interpolation conditions (2.24).

The uniqueness here is within the specified coefficient space, with the fixed node-zero data and constants. Triangular-shell values and first derivatives do not determine an arbitrary radial Schwartz function and its Fourier transform uniquely [8, Theorem 1.7].

Proof. Let p_i^0 denote (2.11) with the fixed values $(C_i, c_{i,0}, d_{i,0})$ and all positive-node coefficients zero. Let $e \in X$ have its d_1 coordinate equal to one and all other coordinates zero. Define $g_i \in X$ by

$$(g_i)_{c,n} = -\frac{K_{p_{3-i}^0}(n)}{Q_n}, \quad (g_i)_{d,n} = \frac{D_n K_{p_{3-i}^0}(n) - K'_{p_{3-i}^0}(n)}{Q_n} - \mathbf{1}_{\{i=1\}} e_{d,n}.$$

The estimates for the node-zero and constant columns show that these are summable lists. The equations to solve are precisely

$$x_1 - Sx_2 = g_1, \quad x_2 - Sx_1 = g_2.$$

Their sum is an equation for $I - S$, and their difference is an equation for $I + S$. Lemma 3.3 gives unique real solutions and hence unique x_1, x_2 .

It remains to locate these solutions relative to the table. Set

$$\rho_i = \tilde{x}_i - S\tilde{x}_{3-i} - g_i.$$

There are 102 scalar coordinates in I_f , so (3.11) gives

$$\|(\rho_i)_f\|_1 < 1.1 \cdot 10^{-7} =: R_f.$$

On the tail, both the tabulated positive-node list and the vector e vanish. The density-column estimate, now including node zero, and (3.9) imply

$$\|(\rho_i)_t\|_1 < 2.29 \cdot 5 \cdot 10^{-10} + .017 \cdot 3.2 \cdot 10^{-8} =: R_t.$$

Here 2.29 bounds the whole tabulated coefficient list, and .017 bounds both fixed constants in absolute value.

The errors $x_i - \tilde{x}_i$ are recovered by taking half the sum and half the difference of the solutions with residuals $-(\rho_1 + \rho_2)$ and $-(\rho_1 - \rho_2)$. Applying (3.12) to both gives, for each list,

$$\|x_i - \tilde{x}_i\|_1 \leq 2(91R_f + 2540R_t) = .00002860012 < .00003.$$

This proves (3.15). Since the fixed node-zero coefficients agree, the total exact coefficient norm is less than $2.29 + .00003 < 2.3$. Finally, (2.25) is equivalent to (2.24) by Lemma 2.4. $\square$

4 Global signs

We now prove the global signs for the exact functions supplied by Proposition 3.4. The strict bounds will also identify their zeros within the respective sign domains. Write

$$\sigma_1 = -1, \quad \sigma_2 = 1, \quad \delta = 3 \cdot 10^{-5}.$$

The difference between each exact coefficient list and its tabulated list has norm less than δ . Their fixed coefficients at zero and their constants C_i agree. Tildes will always denote the functions constructed from the finite tabulated lists, so that $\tilde{H}_i = \tilde{p}_i + K_{\tilde{p}_{3-i}}$. We use the spectral envelopes of Section 3 throughout.

4.1 Taylor quotients on half-gaps

Set $\mathcal{N}_0 = \{0\} \cup \mathcal{N}$. For each $m \in \mathcal{N}_0 \cap [0, 40]$, let m^+ be its successor in $\mathcal{N}_0$ and include

$$y = \frac{m^+ - m}{2}.$$

If $m > 0$, also include $y = (m^- - m)/2$, where m^- is its predecessor. Thus y is signed, and the associated half-gap is

$$I_{m,y} = \{m + yv : 0 \leq v \leq 1\}.$$

For function index 2 we use every such interval, with order $\nu = 0$ at $m = 0$ and $\nu = 2$ otherwise. For function index 1 we use only those intervals contained in $[1, \infty)$, with $\nu = 1$ at $m = 1$ and $\nu = 2$ otherwise. In particular, the only odd order occurs on the rightward half-gap from 1. There are 22 centers from 0 through 40. They give 43 half-gaps for index 2 and, after removing the two intervals not contained in $[1, \infty)$, 41 for index 1, for a total of 84 certified intervals.

Fix one of the specified triples (i, m, y) . For $s = m + u$ in its interval, we will prove positivity of

$$\frac{\sigma_i H_i(m + u)}{u^\nu}.$$

Here $u^0 = 1$, and for $\nu > 0$ the quotient at $u = 0$ is interpreted by continuity. The interpolation conditions make the derivatives of H_i of orders below ν vanish at m , so this extension exists. Positivity of the quotient gives the required sign: every order is even except for the rightward half-gap from 1, where $u \geq 0$. Figure 1 shows the node-to-midpoint intervals on which these quotients are studied.

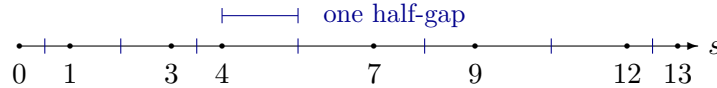


Figure 1: Interpolation nodes (dots) and adjacent gap midpoints (short marks) in the squared-radius coordinate. Each node-to-midpoint segment is a half-gap. The finite sign certificates control a Taylor quotient on each whole half-gap; a separate argument covers the remaining unbounded region.

The tabulated function $\tilde{H}_i$ approximates H_i but need not reproduce its exact zero and slope at a node. We therefore compare the exact quotient with the Taylor series of $\tilde{H}_i$ after deleting the terms

of orders below ν . Put $\Delta_i = H_i - \tilde{H}_i$. Since $\tilde{H}_i$ is entire and the lower derivatives of H_i vanish, this Taylor series gives

$$\begin{aligned} \frac{\sigma_i H_i(m+u)}{u^\nu} - \sigma_i \sum_{r=0}^{\infty} \frac{\tilde{H}_i^{(r+\nu)}(m)}{(r+\nu)!} u^r \\ = \frac{\sigma_i}{u^\nu} \left(\Delta_i(m+u) - \sum_{j=0}^{\nu-1} \frac{\Delta_i^{(j)}(m)}{j!} u^j \right). \end{aligned} \quad (4.1)$$

For $\nu = 0$, the sum on the second line is empty and the right-hand side is simply $\sigma_i \Delta_i(m+u)$. For $\nu > 0$, Taylor's integral remainder gives the exact identity

$$\frac{\Delta_i(m+u) - \sum_{j=0}^{\nu-1} \Delta_i^{(j)}(m) u^j / j!}{u^\nu} = \frac{1}{(\nu-1)!} \int_0^1 (1-t)^{\nu-1} \Delta_i^{(\nu)}(m+tu) dt.$$

It holds for either sign of u and extends continuously to $u = 0$. Since $m+tu$ stays in the same half-gap, the absolute value of the right-hand side of (4.1) is at most

$$\frac{1}{\nu!} \sup_{s \in I_{m,y}} \left| \Delta_i^{(\nu)}(s) \right|. \quad (4.2)$$

The same bound applies at $u = 0$ by continuity and for either orientation of the half-gap. Deleting the lower Taylor terms is essential here: dividing the raw table error by u^ν could be singular at the node.

After writing $u = yv$, the first 29 terms of the tabulated series in (4.1) form a polynomial of degree at most 28 in v . We bound it on $[0, 1]$ using the standard range enclosure by Bernstein coefficients [9, Section 3], retaining its elementary proof below.

Lemma 4.1 (Finite Bernstein certificate). *For every specified triple (i, m, y) , put*

$$T_r = \sigma_i y^r \frac{\tilde{H}_i^{(r+\nu)}(m)}{(r+\nu)!}, \quad 0 \leq r \leq 28. \quad (4.3)$$

Then, for $0 \leq k \leq 28$,

$$B_k := \sum_{r=0}^k \frac{\binom{k}{r}}{\binom{28}{r}} T_r > \begin{cases} 0.74, & m = 0, \\ 0.18, & m = 1, \\ 0.02, & m = 3, 4, \\ 0.009, & 4 < m \leq 40. \end{cases} \quad (4.4)$$

Proof. The primary interval calculation in Appendix A encloses the exact derivatives of the tabulated functions in (4.3), using the quadrature assertion of Lemma A.1 for the analytic moment remainder. Its exact grouped bounds (A.16) imply the stated thresholds. $\square$

The relevance of this finite statement is the identity

$$\sum_{r=0}^{28} T_r v^r = \sum_{k=0}^{28} B_k \binom{28}{k} v^k (1-v)^{28-k}. \quad (4.5)$$

Indeed, for $0 \leq r \leq k \leq d = 28$, the binomial identity

$$\frac{\binom{k}{r}}{\binom{d}{r}} \binom{d}{k} = \binom{d-r}{k-r}$$

gives

$$\sum_{k=r}^d \frac{\binom{k}{r}}{\binom{d}{r}} \binom{d}{k} v^k (1-v)^{d-k} = v^r \sum_{l=0}^{d-r} \binom{d-r}{l} v^l (1-v)^{d-r-l} = v^r.$$

Multiply by T_r , sum over r , and interchange the two finite sums to obtain (4.5). For $0 \leq v \leq 1$, the basis functions on the right are nonnegative and sum to one. Hence the lower bounds in (4.4) hold on the whole interval for the polynomial on the left. To obtain a lower bound for the exact quotient, it remains to control the coefficient error in (4.2) and the terms beyond degree 28 in the tabulated series.

4.2 Bounds for the two errors

The density bounds and (3.6), together with the separate coefficient error bounds δ for the two functions, give, for $s \geq s_* \geq 0$,

$$\left| \Delta_i^{(\nu)}(s) \right| \leq \delta \sum_{j=1}^6 M_j \left(\frac{(\pi t_j)^\nu}{6} + L_j(s_*) (\pi a_{j-1}^*)^\nu \right). \quad (4.6)$$

There are no atom contributions because the constants C_i have not changed. Substitution of the envelopes proves the following uniform bounds for (4.2):

$$\frac{1}{\nu!} \sup_{I_{m,y}} \left| \Delta_i^{(\nu)} \right| < \begin{cases} 0.004, & m = 0, \\ 0.011, & m = 1, \\ 0.002, & m \geq 3. \end{cases} \quad (4.7)$$

Here one uses $s_* = 0$ for $m = 0$, $s_* = 1/2$ for $m = 1$ and both possible orders, and $s_* = 2$ for $m \geq 3$. For example the first bound is $\delta \sum_{j=1}^6 (M_j/6 + M_j L_j(0)) < 0.004$. All these are bounds on intervals, obtained from the derivative envelopes.

It remains to truncate the infinite series in (4.1). Put $Y = |y|$ and $q = 29 + \nu$. The total variation of the direct spectral measure of either tabulated function is less than 80, since

$$2.29 \sum_{j=1}^6 \frac{M_j}{6} + 0.017 \sum_{j=0}^6 m_j^* < 80.$$

For the transformed part, use coefficient norm less than 2.3 and $|C_i| \leq 0.017$. Expanding each spectral exponential at m gives the following bound for the omitted quotient terms:

$$\begin{aligned} \mathcal{E}(m, Y, \nu) := & \frac{Y^{-\nu}}{q!} \frac{1}{1 - \pi a_0^* Y / (q+1)} \left(80(\pi Y)^q \right. \\ & \left. + 2.3 \sum_{j=1}^6 M_j L_j(m) (\pi a_{j-1}^* Y)^q + 0.017 \sum_{j=0}^6 L_j^P(m) (\pi a_j^* Y)^q \right). \end{aligned} \quad (4.8)$$

Indeed the first omitted exponential term has degree q and subsequent absolute-value ratios are at most $\pi a_0^* Y / (q+1) < 1$. After division by u^ν , all powers still have degree at least 29, so their maximum for $|u| \leq Y$ is bounded by evaluating these absolute-value majorants at Y . This proves (4.8) also when $y < 0$.

The five representative substitutions are

$$(m, Y, \nu) = (0, \tfrac{1}{2}, 0), (1, 1, 1), (1, 1, 2), (3, 1, 2), (4, \tfrac{3}{2}, 2). \quad (4.9)$$

In each case (4.8) is less than 0.001. Increasing m or decreasing Y only decreases the bound. The first three cases handle centers 0 and 1, the fourth handles $m = 3$, and the last handles every center $m \geq 4$. Thus the truncation error is less than 0.001 on every required half-gap.

Combining (4.5), (4.1), (4.7), and (4.8), we obtain

$$\frac{\sigma_i H_i(s)}{(s-m)^\nu} > \begin{cases} 0.735, & m = 0, \\ 0.168, & m = 1, \\ 0.017, & m = 3, 4, \\ 0.006, & 4 < m \leq 40, \end{cases} \quad s \in I_{m,y}, \quad (4.10)$$

with endpoint values interpreted by continuity. Since the only odd order has $s - m \geq 0$, these inequalities imply the desired signs on all of these intervals. Their union is $[0, 41.5]$ for index 2 and $[1, 41.5]$ for index 1: adjacent half-gaps meet at their common midpoint, and the last interval runs from 40 halfway to 43.

4.3 The unbounded region

For $s \geq s_0 := 41.5$, split the exact direct part as

$$\begin{aligned} p_i(s) &= P(s)R_{i,0}(s) + p_{i,1}(s), \\ R_{i,0}(s) &= C_i + \sum_{\substack{n \in \mathcal{N}_0 \\ n \leq 40}} \left(\frac{c_{i,n}}{(s-n)^2} + \frac{d_{i,n}}{s-n} \right), \\ \tilde{R}_{i,0}(s) &= C_i + \sum_{\substack{n \in \mathcal{N}_0 \\ n \leq 40}} \left(\frac{\tilde{c}_{i,n}}{(s-n)^2} + \frac{\tilde{d}_{i,n}}{s-n} \right). \end{aligned} \quad (4.11)$$

Here $p_{i,1}$ is the cardinal contribution from exact coefficients at nodes greater than 40. The function $R_{i,0}$ uses the exact coefficients, whereas $\tilde{R}_{i,0}$ uses the finite table; their constants and node-zero coefficients agree. All denominators in these two rational functions are at least 1.5. Write the corresponding decomposition of the exact function as

$$E_i := p_{i,1} + K_{p_{3-i}}, \quad H_i = PR_{i,0} + E_i.$$

For each $s \geq s_0$, the final comparison will use a nearest node $m \geq 43$. At such a node, the exact interpolation conditions and the double zero of P make E_i have zero value and slope; we verify this at the comparison step below. The useful scale for both terms is therefore $(s-m)^2$. We will bound $\sigma_i R_{i,0}$ below, bound $P(s)$ below by a multiple of $(s-m)^2$, and bound $|E_i(s)|$ above at the same scale through a uniform second-derivative estimate.

We first show

$$\sigma_i R_{i,0}(s) > 0.011 \quad (s \geq s_0, i = 1, 2). \quad (4.12)$$

For the tabulated coefficients put $u_n^* = \sigma_i \tilde{c}_{i,n}$, $v_n^* = \sigma_i \tilde{d}_{i,n}$, and $[w]_- := \min\{w, 0\}$. Using

$$\frac{1}{s-n} = \frac{1}{s} + \frac{n}{s(s-n)}$$

and discarding positive terms gives the following lower bound for $\sigma_i \tilde{R}_{i,0}(s)$, uniformly for $s \geq s_0$:

$$\sigma_i C_i + \frac{[\sum_{n \leq 40} v_n^*]_-}{s_0} + \sum_{n \leq 40} \left(\frac{[u_n^*]_-}{(s_0 - n)^2} + \frac{[v_n^*]_- n}{s_0(s_0 - n)} \right), \quad (4.13)$$

where both sums include $n = 0$. Each negative reciprocal term is smallest at $s = s_0$. The exact rational sums in (4.13) exceed 0.012 for index 1 and 0.016 for index 2. To pass to the exact rational part, use $s - n \geq 1.5 > 1$ and the coefficient enclosure to obtain

$$\begin{aligned} |R_{i,0}(s) - \tilde{R}_{i,0}(s)| &\leq \sum_{n \leq 40} \left(\frac{|c_{i,n} - \tilde{c}_{i,n}|}{(s-n)^2} + \frac{|d_{i,n} - \tilde{d}_{i,n}|}{s-n} \right) \\ &< \frac{\delta}{s_0 - 40} = 0.00002. \end{aligned}$$

The finite lower sums therefore imply the asserted exact bound (4.12).

The coefficient norm of the tabulated part with $n > 40$ is less than 0.00002 for either function. Hence the exact coefficient norm for $p_{i,1}$ is less than 0.00005. The derivative envelopes imply

$$\begin{aligned} \sup_{s \geq s_0} |E_i''(s)| &\leq 0.00005 \cdot 77 + 2.3 \sum_{j=1}^6 M_j L_j(s_0) (\pi a_{j-1}^*)^2 \\ &\quad + 0.017 \sum_{j=0}^6 L_j^P(s_0) (\pi a_j^*)^2 < 0.006. \end{aligned} \tag{4.14}$$

For the direct part we used $\sum_{j=1}^6 M_j (\pi t_j)^2 / 6 < 77$; for the transformed part we used the all-node coefficient norm less than 2.3 from Proposition 3.4.

It remains to give the quadratic lower bound for P .

Lemma 4.2. *If $m \in 12\mathbb{Z} + A$ is a nearest zero of P to a real number s , then*

$$P(s) \geq 0.68(s - m)^2.$$

Proof. On either half-gap from m , consider $P(m + u)/u^2$, with value Q_m at $u = 0$. Put $a = \pi/12$. For a sine factor whose residue is $a' \neq m$ modulo 12, the second derivative of its logarithm is $-2a^2 \csc^2(a(m + u - a')) < 0$. These factors stay nonzero on the half-gap. The factor vanishing at m , after division by u^2 , has the same logarithmic concavity because

$$\frac{d^2}{du^2} \log \left| \frac{\sin(au)}{u} \right| = \frac{1}{u^2} - \frac{a^2}{\sin^2(au)} \leq 0.$$

Here $|\sin(au)| \leq |au|$, and the logarithm has a smooth limiting value at $u = 0$. Squaring the divided sine only multiplies this second derivative by two. The quotient is therefore logarithmically concave on each half-gap, including its continuous value at the center. Its minimum is bounded below by the smaller of its endpoint values. At the center $Q_m > 1.75$. At the gap midpoints, direct substitution gives the values listed below; periodicity reduces the calculation to one period. To evaluate them, put $\theta = \pi(s - 2)/12$ and $c = \cos(2\theta)$. The identity $4 \sin v \sin w = 2(\cos(v - w) - \cos(v + w))$ shows that the paired unsquared factors for residues (0, 4), (1, 3), and (7, 9) are, respectively,

$$1 - 2c, \quad \sqrt{3} - 2c, \quad \sqrt{3} + 2c.$$

Consequently

$$P(s) = (4c^2 - 3)^2(1 - 2c)^2, \quad c = \cos \frac{\pi(s - 2)}{6},$$

which evaluates every entry directly.

Gap between consecutive zeros	Midpoint quotient $P(s)/(s-m)^2$
$[0, 1], [3, 4]$	$12 - 8\sqrt{2}$
$[1, 3]$	1
$[7, 9]$	9
$[4, 7], [9, 12]$	$\frac{8}{9}(3 + \sqrt{2} + \sqrt{3} + \sqrt{6})$

The distances to the two ends of a gap agree at its midpoint, so this table supplies the endpoint value for both half-gaps. The smallest value is $12 - 8\sqrt{2} > 0.68$: indeed $(283/200)^2 > 2$, so $\sqrt{2} < 283/200$ and $12 - 8(283/200) = 0.68$. The other three values are at least 1. This proves the claim. $\square$

For any $s \geq 41.5$, choose a nearest node $m \geq 43$; at $s = 41.5$, choose 43 rather than the tied node 40. Both H_i and $PR_{i,0}$ vanish to order at least two at m , by (2.24) and the absence of a pole of $R_{i,0}$ there. Thus $E_i = H_i - PR_{i,0}$ vanishes to order at least two at m . The segment between s and m lies in $[41.5, \infty)$, and (4.14) gives

$$|E_i(s)| \leq \frac{0.006}{2}(s-m)^2.$$

Using (4.12) and Lemma 4.2, we obtain on the entire unbounded region

$$\sigma_i H_i(s) \geq \left(0.011 \cdot 0.68 - \frac{0.006}{2}\right)(s-m)^2 = 0.00448(s-m)^2 \geq 0. \quad (4.15)$$

Proposition 4.3 (Global signs and zeros). *The exact functions have the zero sets*

$$\{s \geq 1 : H_1(s) = 0\} = \mathcal{N}, \quad \{s \geq 0 : H_2(s) = 0\} = \mathcal{N}, \quad (4.16)$$

and satisfy the strict inequalities

$$H_1(s) < 0 \quad (s \geq 1, s \notin \mathcal{N}), \quad H_2(s) > 0 \quad (s \geq 0, s \notin \mathcal{N}).$$

The zero of H_1 at 1 is simple. Its zeros at $n \in \mathcal{N}$ with $n > 1$, and the zeros of H_2 at every $n \in \mathcal{N}$, have order exactly two as zeros of the entire functions of s . Consequently $F(x) \leq 0$ when $b|x|^2 \geq 1$, and $\widehat{F}(\xi) \geq 0$ for every $\xi \in \mathbb{R}^2$.

Proof. At every node $n \in \mathcal{N}$, the exact interpolation conditions give $H_1(n) = H_2(n) = 0$. Away from the nodes, the quotient bounds (4.10) are strict on the finite region, and (4.15) is strict on the remaining region. This proves (4.16) and the asserted strict signs.

The interpolation slope $H_1'(1) = -Q_1 \neq 0$ makes its zero at 1 simple. For H_2 at every node in $\mathcal{N} \cap [1, 40]$, and for H_1 at those nodes greater than 1, the chosen order is $\nu = 2$. The endpoint value in (4.10) is then $\sigma_i H_i''(m)/2 > 0$, so the zero has order exactly two. For a node $m \in \mathcal{N}$ with $m \geq 43$, take u sufficiently small that m remains the nearest node to $m+u$ and $m+u \geq 41.5$. The exact value and first derivative at m vanish, so (4.15) yields

$$\frac{\sigma_i H_i''(m)}{2} = \lim_{u \rightarrow 0} \frac{\sigma_i H_i(m+u)}{u^2} \geq 0.00448 > 0.$$

These zeros also have order exactly two. Finally, the factors $e^{-\pi h s}$ in F and $\widehat{F}$ are positive, which gives their non-strict sign inequalities. $\square$

5 Poisson normalization

The prescribed first-shell derivative determines the remaining scalar normalization. Poisson summation first equates the values at the origin; its differentiated dilation identity then proves that their common value is positive, without estimating that value numerically.

Let the exact coefficient lists be those of Proposition 3.4, and put

$$G_i(s) = e^{-\pi h s} H_i(s), \quad F(x) = G_1(b|x|^2).$$

Section 2 proves that F is real, radial, and Schwartz, with $\widehat{F}(\xi) = G_2(b|\xi|^2)$. The interpolation conditions give

$$\begin{aligned} G_1(n) &= G_2(n) = 0, \\ G'_2(n) &= 0, \quad G'_1(n) = -Q_1 e^{-\pi h} \mathbf{1}_{\{n=1\}} \quad (n \in \mathcal{N}). \end{aligned} \tag{5.1}$$

Indeed, $G'_i(n) = e^{-\pi h n} H'_i(n)$ because $H_i(n) = 0$.

5.1 The dilation identity

Use the lattice Λ of covolume one and the shell coordinate s_a from (2.2)–(2.3). Every positive s_a lies in $\mathcal{N}$. Exactly six lattice vectors have $s_a = 1$: from $4(j^2 + jk + k^2) = (2j + k)^2 + 3k^2 = 4$, one obtains

$$(j, k) = (1, 0), (-1, 0), (0, 1), (0, -1), (1, -1), (-1, 1).$$

As shown in Section 2, $\Lambda^* = \mathcal{R}\Lambda$. Apply Poisson summation on Λ to $F_t(x) = F(\sqrt{t}x)$, whose Fourier transform is $t^{-1}\widehat{F}(\xi/\sqrt{t})$. Radiality makes the sums over Λ^* and Λ equal, and gives, for every $t > 0$,

$$\sum_{a \in \Lambda} G_1(ts_a) = \frac{1}{t} \sum_{a \in \Lambda} G_2(s_a/t). \tag{5.2}$$

We will differentiate this identity twice. Here is the needed convergence justification. On a compact subinterval of $(0, \infty)$, each of the first two t derivatives of $F(\sqrt{t}x)$ is a finite sum of spatial derivatives of F multiplied by polynomials in x and bounded functions of t . The same statement holds for $t^{-1}\widehat{F}(x/\sqrt{t})$. The Schwartz bounds therefore imply, for every M , a common bound $C_M(1 + |x|)^{-M}$ for these functions and their first two t derivatives. The number of points of a fixed planar lattice in a radius- R disk is $O(R^2)$, so $\sum_{a \in \Lambda} (1 + |a|)^{-M}$ converges for $M > 2$. Uniform convergence on compact t intervals now permits both termwise differentiations of (5.2).

Lemma 5.1 (Exact origin value). *The Fourier pair constructed above satisfies*

$$F(0) = \widehat{F}(0) = 6Q_1 e^{-\pi h} > 0. \tag{5.3}$$

Proof. At $t = 1$, all nonzero summands in (5.2) vanish by (5.1). Hence $G_1(0) = G_2(0)$. The derivative of its left side at 1 is

$$\sum_{a \in \Lambda} s_a G'_1(s_a) = -6Q_1 e^{-\pi h},$$

because only the six vectors on the first shell contribute. On the right,

$$\left. \frac{d}{dt} (t^{-1} G_2(s/t)) \right|_{t=1} = -G_2(s) - s G'_2(s).$$

For $s = s_a > 0$, both terms vanish by the exact jets, while $s_a = 0$ contributes $-G_2(0)$. Equality of the derivatives gives $G_2(0) = 6Q_1 e^{-\pi h}$. The positivity follows from $Q_1 > 0$ in (2.9). This calculation uses the Fourier pairing and the exact interpolation conditions, but not the global sign estimates. $\square$

5.2 The final dilation

Proof of Theorem 1.1. Define

$$f(x) = \frac{F(x/\sqrt{b})}{bF(0)}, \quad \widehat{f}(\xi) = \frac{\widehat{F}(\sqrt{b}\xi)}{F(0)}. \quad (5.4)$$

The second identity is the two-dimensional Fourier scaling rule. Since $F(0) > 0$, the dilation and normalization preserve real-valuedness, radially, and Schwartz regularity. They give $\widehat{f}(0) = 1$ and $f(0) = 1/b = 2/\sqrt{3}$. By Proposition 4.3 and the positive factor $e^{-\pi hs}$, $G_2(s) = e^{-\pi hs}H_2(s) \geq 0$ for $s \geq 0$ and $G_1(s) = e^{-\pi hs}H_1(s) \leq 0$ for $s \geq 1$. Thus $\widehat{f} \geq 0$ everywhere. If $|x| \geq 1$, then $b|x/\sqrt{b}|^2 = |x|^2 \geq 1$, so $f(x) \leq 0$. These are exactly the four assertions of (1.1). $\square$

The scaling also records the prescribed contact data in the final radial coordinates. Write $f_{\text{rad}}(r) = f(r, 0)$ and $\widehat{f}_{\text{rad}}(\rho) = \widehat{f}(\rho, 0)$ for $r, \rho \geq 0$. Then

$$f_{\text{rad}}(r) = \frac{G_1(r^2)}{bF(0)}, \quad \widehat{f}_{\text{rad}}(\rho) = \frac{G_2(b^2\rho^2)}{F(0)}.$$

Proposition 4.3 therefore gives the exact zero sets in the respective sign domains:

$$\begin{aligned} \{r^2 : r \geq 1, f_{\text{rad}}(r) = 0\} &= \mathcal{N}, \\ \{\rho^2 : \rho \geq 0, \widehat{f}_{\text{rad}}(\rho) = 0\} &= b^{-2}\mathcal{N} = \frac{4}{3}\mathcal{N}. \end{aligned} \quad (5.5)$$

The physical zero at $r = 1$ is simple, every other listed physical zero is exactly double, and every listed Fourier zero is exactly double. Indeed, the Gaussian and scalar factors do not vanish, while the changes of variable $s = r^2$ and $s = b^2\rho^2$ have nonzero derivatives at these positive radii, so they preserve the orders established for H_1, H_2 . At the exclusion radius, the radial derivative is

$$f'_{\text{rad}}(1) = \frac{2G'_1(1)}{bF(0)} = -\frac{1}{3b} = -\frac{2}{3\sqrt{3}}. \quad (5.6)$$

5.3 The packing consequence

The relation between the signs and packing density is particularly visible for a periodic packing. Suppose the center set is $\bigcup_{j=1}^N (L + t_j)$, where L has covolume V , the t_j are distinct modulo L , and distinct centers have distance at least one. Absolute convergence follows from the Schwartz bounds. Every summand below except the N diagonal terms at the origin is nonpositive, so

$$\mathcal{S} := \sum_{j,k=1}^N \sum_{v \in L} f(v + t_j - t_k) \leq Nf(0).$$

Poisson summation on each translated lattice gives

$$\mathcal{S} = \frac{1}{V} \sum_{w \in L^*} \widehat{f}(w) \left| \sum_{j=1}^N e^{2\pi i w \cdot t_j} \right|^2 \geq \frac{N^2}{V} \widehat{f}(0),$$

because every Fourier summand is nonnegative. It follows that $N/V \leq f(0)/\widehat{f}(0) = 2/\sqrt{3}$. The disks have area $\pi/4$, hence their covered-area density is at most $\pi/(2\sqrt{3})$.

For an arbitrary, not necessarily periodic, packing, the same implication is Theorem 3.1 of Cohn and Elkies [2]. Its admissibility hypotheses hold because both f and $\hat{f}$ are Schwartz. For disks of radius $1/2$, the Cohn–Elkies center-density normalization is one quarter of the center-number density in their upper-density convention. Multiplying it by the unit-disk area π therefore gives the corresponding covered-area density, equivalently $\pi/4$ times that center-number density. Their upper density takes a supremum over observation centers, so it bounds the fixed-origin limsup used in the introduction. For the fixed-origin convention and separated centers, the covered area inside B_R lies between $(\pi/4)$ times the number of centers in $B_{R-1/2}$ and $(\pi/4)$ times the number in $B_{R+1/2}$; division by πR^2 and passage to the limsup matches the convention in the introduction. The lattice generated by $(1, 0)$ and $(1/2, b)$ has one center per area b and attains this value. This is the classical packing-density theorem recovered from the new Fourier certificate, not a new density claim.

The restriction of the physical zero set to integer squared radii also recovers the classical periodic equality case. We use the even integral lattice argument of Cohn and Elkies [2, Section 8].

Corollary 5.2 (Periodic equality case). *If a periodic packing of disks of radius $1/2$ has covered-area density $\pi/(2\sqrt{3})$, then its center set is the image of $\mathbb{Z}(1, 0) + \mathbb{Z}(1/2, b)$ under a Euclidean isometry.*

Proof. Write the center set as $\mathcal{C} = \bigcup_{j=1}^N (L + t_j)$, with the notation of the periodic packing argument above. Equality of its density with $\pi/(2\sqrt{3})$ gives $N/V = f(0)/\hat{f}(0)$. Thus both inequalities in the Poisson comparison are equalities. Absolute convergence permits us to write the defect in the physical inequality as

$$0 = Nf(0) - \mathcal{S} = \sum_{\substack{1 \leq j, k \leq N, \\ v \in L \\ v \neq 0 \text{ or } j \neq k}} -f(v + t_j - t_k).$$

Every summand is nonnegative, so every one is zero. Any difference of distinct centers occurs in this sum, and (5.5) therefore gives

$$|x - y|^2 \in \mathcal{N} \subset \mathbb{Z} \quad (x, y \in \mathcal{C}, x \neq y).$$

Only the physical zero set is used here.

Translate one center to the origin and dilate by $\sqrt{2}$. Put

$$L_0 = \sqrt{2}L, \quad \mathcal{C}_0 = \sqrt{2}(\mathcal{C} - t_1), \quad \Gamma = \text{span}_{\mathbb{Z}} \mathcal{C}_0.$$

Here Γ consists of finite integer linear combinations of the scaled centers, and $L_0 \subset \mathcal{C}_0$ because the first coset becomes L_0 . All squared distances in $\mathcal{C}_0$ are even integers. Since $0 \in \mathcal{C}_0$, polarization gives

$$\langle x, y \rangle = \frac{|x|^2 + |y|^2 - |x - y|^2}{2} \in \mathbb{Z} \quad (x, y \in \mathcal{C}_0).$$

Integer linear combinations preserve these integral pairings, and their squared norms remain even because the cross terms carry a factor of two. Thus every pairing in Γ is integral and every squared norm in Γ is even.

In particular, pairing with a basis of L_0 shows $L_0 \subset \Gamma \subset L_0^*$. Both L_0 and L_0^* are lattices of rank two, so L_0^*/L_0 is finite. Hence Γ is a finite union of L_0 cosets and is itself a lattice of rank two. Put $d := [\Gamma : L_0]$. A basis of Γ has a Gram matrix

$$\begin{pmatrix} a & m \\ m & c \end{pmatrix}, \quad a, c \in 2\mathbb{Z}_{>0}, \quad m \in \mathbb{Z}.$$

Its determinant $ac - m^2 = \text{covol}(\Gamma)^2$ is a positive integer congruent to 0 or 3 modulo 4. It is therefore at least 3, and $\text{covol}(\Gamma) \geq \sqrt{3}$. On the other hand, $\mathcal{C}_0$ consists of N distinct L_0 cosets, all among the d cosets of Γ . Since $\text{covol}(L_0) = 2V$, we obtain

$$\sqrt{3} \leq \text{covol}(\Gamma) = \frac{2V}{d} \leq \frac{2V}{N} = \sqrt{3}.$$

Equality holds throughout, so $d = N$ and the two sets of cosets coincide: $\mathcal{C}_0 = \Gamma$.

It remains to identify this lattice of Gram determinant 3. Choose a shortest nonzero vector $w_1 \in \Gamma$ and extend it to a lattice basis w_1, w_2 ; this is possible because a shortest vector is primitive. Put $a = |w_1|^2$, and replace w_2 by an integer translate along w_1 so that $m = \langle w_1, w_2 \rangle$ satisfies $|m| \leq a/2$. With $c = |w_2|^2 \geq a$ and the even integer $a \geq 2$,

$$3 = ac - m^2 \geq \frac{3}{4}a^2.$$

Hence $a = 2$. The equation $2c - m^2 = 3$, with c even and $|m| \leq 1$, forces $c = 2$ and $m = \pm 1$. This is the Gram matrix of the triangular lattice with shortest length $\sqrt{2}$. Scaling back by $1/\sqrt{2}$ proves the assertion. $\square$

5.4 A second-derivative consistency identity

The same dilation identity contains an additional relation among the exact curvatures. It is not needed for the construction or its sign proof. For $n \in \mathcal{N}$, let $r_\Lambda(n) = \#\{a \in \Lambda : s_a = n\}$, allowing the value zero for nodes that are not lattice radii; in particular, $r_\Lambda(1) = 6$.

Proposition 5.3 (Second differentiated Poisson identity). *The exact pair satisfies the absolutely convergent identity*

$$\sum_{n \in \mathcal{N}} r_\Lambda(n) n^2 (G_1''(n) - G_2''(n)) = 2G_2(0). \quad (5.7)$$

Using the global signs, it follows that

$$H_1''(1) - H_2''(1) \geq 2(1 - \pi h)Q_1. \quad (5.8)$$

Proof. The convergence argument above permits a second differentiation of (5.2). At $t = 1$,

$$\left. \frac{d^2}{dt^2} (t^{-1}G_2(s/t)) \right|_{t=1} = 2G_2(s) + 4sG_2'(s) + s^2G_2''(s).$$

On the left the corresponding term is $s^2G_1''(s)$. At every positive node, the first two terms on the right vanish by (5.1); at the origin they contribute $2G_2(0)$. Grouping the remaining absolutely convergent lattice sums by their squared radii proves (5.7).

For a positive node $n > 1$, G_1 has a local maximum equal to zero and G_2 has a local minimum equal to zero, by Proposition 4.3 and $e^{-\pi h s} > 0$. Therefore $G_1''(n) \leq 0 \leq G_2''(n)$. All terms in (5.7) beyond the first shell are nonpositive. Since $r_\Lambda(1) = 6$ and $G_2(0) = 6Q_1e^{-\pi h}$,

$$G_1''(1) - G_2''(1) \geq 2Q_1e^{-\pi h}.$$

Finally, differentiating $G_i = e^{-\pi h s}H_i$ twice and inserting the exact value and slope at 1 gives

$$G_1''(1) - G_2''(1) = e^{-\pi h} (H_1''(1) - H_2''(1) + 2\pi h Q_1).$$

This yields (5.8). Unlike the origin calculation, this last inequality is conditional on the global signs that were proved in Section 4. $\square$

A Finite certificates

This appendix completes the finite evidence used in Lemma 3.2 and Lemma 4.1. We give the exact inputs, prove a uniform error bound for the finite quadrature sums, and state the interval comparisons for the matrices, residuals, and Bernstein coefficients. The final subsection records the scalar substitutions used for the infinite tails and continuum signs. All terminating decimals denote exact rational numbers.

The primary calculation uses interval evaluation of the $N = 256$ quadrature sums, enlarged by the analytic moment remainder proved below. It therefore reduces the required numerical evidence to bounded finite sums, without assuming convergence of a numerical integration or optimization procedure. The infinite inverse and the signs between and beyond interpolation nodes remain consequences of the analytic arguments in the main text. Appendix B gives the rational alternatives and their separately stated scope.

A.1 Exact input lists and matrices

Table 2 gives the coefficients at every node of $I_f = \mathcal{N} \cap [1, 100]$. Its four coefficient columns are multiplied by 10^{10} : an integer a in the table means the coefficient $a 10^{-10}$. At all nodes $n > 100$ both approximate lists are zero. The data at zero and the constant terms are, exactly,

$$(\tilde{c}_{1,0}, \tilde{d}_{1,0}, C_1) = (1, .44, -.013), \quad (\tilde{c}_{2,0}, \tilde{d}_{2,0}, C_2) = (0, -.368, .017). \quad (\text{A.1})$$

The zero-node data are included in all evaluations of the tabulated functions. They are excluded when forming a column of the operator S . Direct integer addition gives

$$\begin{aligned} \sum_{n \geq 0} (|\tilde{c}_{1,n}| + |\tilde{d}_{1,n}|) &= 2.2516666247 < 2.29, \\ \sum_{n \geq 0} (|\tilde{c}_{2,n}| + |\tilde{d}_{2,n}|) &= .8927477314 < 2.29, \\ \sum_{n > 40} (|\tilde{c}_{1,n}| + |\tilde{d}_{1,n}|) &= .0000197471 < .00002, \\ \sum_{n > 40} (|\tilde{c}_{2,n}| + |\tilde{d}_{2,n}|) &= .0000161623 < .00002. \end{aligned} \quad (\text{A.2})$$

Here and below an index $n \geq 0$ in a coefficient sum ranges over $\{0\} \cup \mathcal{N}$.

Table 2: Rational coefficient approximations. Each of the four coefficient columns is scaled by 10^{10} .

n	$10^{10}\tilde{c}_{1,n}$	$10^{10}\tilde{d}_{1,n}$	$10^{10}\tilde{c}_{2,n}$	$10^{10}\tilde{d}_{2,n}$
1	-670689128	-6398628799	-250018302	4107732082
3	93524468	564897270	-186215974	-65204793
4	-64146977	121133118	82882809	-327822296
7	-1358193	-16329465	1979670	21637038
9	-2932910	-940875	3636920	2145806
12	3646952	-34846939	-6005382	34369042
13	3183411	63030995	-574123	-75237042
15	11390977	27049982	-11312377	-34985096

Continued on the next page

Table 2 (continued)

n	$10^{10}\widetilde{c}_{1,n}$	$10^{10}\widetilde{d}_{1,n}$	$10^{10}\widetilde{c}_{2,n}$	$10^{10}\widetilde{d}_{2,n}$
16	-5593246	13405071	5915710	-10980954
19	54905	-1515007	-120136	1498351
21	-210208	-634307	177992	747946
24	976077	-804949	-1037705	105643
25	-824472	7003719	1011732	-6510015
27	611891	4536095	-428528	-4440927
28	-472882	-107613	399655	403751
31	30744	-112653	-31773	90515
33	-4974	-105568	1240	99081
36	128313	246018	-114876	-267388
37	-176685	505187	166591	-396209
39	-21638	472037	32128	-402283
40	-18461	-165597	10836	164974
43	4565	-2296	-4050	547
45	1477	-10734	-1524	8978
48	10748	50050	-8611	-45048
49	-20506	7495	17354	-266
51	-10361	29035	9641	-21534
52	2121	-26523	-2269	23159
55	425	858	-344	-839
57	302	-632	-266	463
60	382	5658	-231	-4688
61	-1481	-4320	1140	4098
63	-1494	-352	1265	647
64	558	-2609	-491	2114
67	21	169	-14	-145
69	34	13	-28	-18
72	-55	394	53	-301
73	-21	-805	5	685
75	-136	-379	108	338
76	70	-144	-58	104
79	-1	19	1	-15
81	2	9	-2	-8
84	-13	3	11	0
85	13	-87	-11	70
87	-6	-60	4	49
88	6	5	-4	-5
91	0	1	0	-1
93	0	1	0	-1
96	-2	-4	1	3
97	2	-6	-2	4
99	0	-6	0	5
100	0	2	0	-2

Write $J = \{1, 3, 4\}$ and $E = I_f \setminus J$. Matrix rows and columns on a node set always list all c entries first and all d entries second, with nodes increasing within each group. In particular the

order on J is $(c_1, c_3, c_4, d_1, d_3, d_4)$. The two rational matrices used to certify the small inverses are

$$W_+ = 10^{-4} \begin{pmatrix} 4159 & 4452 & -2003 & -10467 & -2421 & -1683 \\ 2714 & 6573 & 1689 & 8258 & 104 & -335 \\ -4675 & 4785 & 7003 & -13100 & -1355 & -1163 \\ 36853 & -37142 & 27514 & 105973 & 1100 & 2012 \\ 57838 & -56941 & 36296 & 158623 & 29097 & 17293 \\ -4705 & 3589 & -3311 & -11932 & -2562 & 6919 \end{pmatrix}, \quad (\text{A.3})$$

$$W_- = 10^{-4} \begin{pmatrix} 13265 & -1358 & -385 & -381 & 815 & -164 \\ 616 & 10426 & 686 & -484 & 182 & 775 \\ -146 & 4 & 9842 & 215 & 267 & 179 \\ -2761 & 2483 & -958 & 5174 & -328 & 206 \\ -615 & -861 & 730 & -1388 & 5619 & -4370 \\ 874 & 820 & 1148 & -431 & 905 & 11927 \end{pmatrix}. \quad (\text{A.4})$$

These matrices are exact inputs. The W -based inverse argument below uses their products and column sums; it does not require a computed matrix inverse. We describe the supplied program's additional inverse checks in Section A.3.

A.2 Moments and the integration rule

For a finite input $p = PR$ supported on nodes at most 100, put

$$U_{p,l}(m, y) = \frac{y^l p^{(l)}(m)}{l!}, \quad X_{p,l}(m, y) = \frac{y^l K_p^{(l)}(m)}{l!}. \quad (\text{A.5})$$

The domain for the quadrature error estimate is stated explicitly in Lemma A.1.

We use the first Fejér quadrature rule, whose cosine-sum weights are described in [11, Section 2, formula (2.3)]. Our weights are half the mass-two weights there, normalized to average over an interval; for even $N = 256$ the terminal cosine term in that formula vanishes. We prove the exactness, positivity and error bound needed here. Set $N = 256$, and for $0 \leq v < N$ define

$$\begin{aligned} \alpha_v &= \frac{v + 1/2}{N}, & o_v &= \frac{1}{N} \left(1 - 2 \sum_{a=1}^{127} \frac{\cos(2a\pi\alpha_v)}{4a^2 - 1} \right), \\ \tau_{jv} &= \frac{2j - 1 + \cos(\pi\alpha_v)}{12} & (1 \leq j \leq 6). \end{aligned} \quad (\text{A.6})$$

Integration over (t_{j-1}, t_j) is replaced by evaluation at τ_{jv} with weights $o_v/6$. To check exactness, expand a polynomial in the Chebyshev basis $T_k(\cos \theta) = \cos(k\theta)$. Discrete cosine orthogonality at $\theta_v = \pi\alpha_v$ recovers its coefficients through degree 255. The integral averages of this basis on $[-1, 1]$ are zero at odd k and $1/(1 - k^2)$ at even k . These are exactly the coefficients in (A.6), so the rule is exact through degree 255. Moreover,

$$o_v \geq \frac{1}{256} \left(1 - \sum_{a=1}^{127} \frac{2}{4a^2 - 1} \right) = \frac{1}{256 \cdot 255} > 0, \quad \sum_v o_v = 1.$$

Thus the six pieces together have total weight one.

For clarity, the complete density on piece j used in this appendix is

$$f_j(t) = 2\pi \sum_n e^{-i\pi t n} \left\{ -\pi c_n \sum_{l=j}^6 (t_l - t) P_l e^{i\pi t_l n} + i d_n \sum_{l=j}^6 P_l e^{i\pi t_l n} \right\}. \quad (\text{A.7})$$

It includes node 0 when that coefficient is part of the input. The folded atoms have masses $e_j CP_j$ at t_j , where $e_0 = 1$ and $e_j = 2$ for $j > 0$. The factors multiplying this density and these masses for U and X , respectively, are

$$e^{i\pi t m} \frac{(i\pi t y)^l}{l!}, \quad \lambda(t) e^{i\pi z(t)m} \frac{(i\pi z(t)y)^l}{l!}. \quad (\text{A.8})$$

The real part is taken after summation.

Lemma A.1 (Quadrature moment error). *Let $p = PR$ have a real coefficient list supported in $\{0\} \cup I_f$, with $\|(c, d)\|_1 \leq 3$, and let $C \in \mathbb{R}$ satisfy $|C| \leq 1$. Let $m, y \in \mathbb{R}$ and $l \in \{0, \dots, 30\}$ satisfy $0 \leq m \leq 100$ and $|y| \leq 3/2$. The quadrature rule (A.6), with exact evaluations at its nodes and at the atoms, has error less than 10^{-22} in each moment $U_{p,l}(m, y)$ and $X_{p,l}(m, y)$.*

This domain contains all basis inputs and the two tabulated inputs used below.

Proof. Put $\rho = 1/6$, $c_j = (2j - 1)/12$, $a_j = c_j + ih$, and $\Delta_j = |a_j|^2 - \rho^2$. These last quantities increase from $\Delta_1 = 167/1200$ to $\Delta_6 = 389/400$, so the disk $|t - c_j| \leq \rho$ does not contain the pole $-ih$. All factors in (A.7) and (A.8) are analytic there. The reciprocal-disk calculation is exact: for $\zeta = 1/(t + ih)$,

$$|1 - a_j \zeta|^2 - \rho^2 |\zeta|^2 = \Delta_j \left| \zeta - \frac{\bar{a}_j}{\Delta_j} \right|^2 - \frac{\rho^2}{\Delta_j}.$$

Thus inversion maps the disk for $t + ih$ onto the disk with center $(c_j - ih)/\Delta_j$ and radius ρ/Δ_j . In particular,

$$|\zeta| \leq \frac{|a_j| + \rho}{\Delta_j} = \frac{1}{|a_j| - \rho}.$$

The right side decreases with c_j , and $|a_1| = \sqrt{601}/60$. Consequently the following are analytic bounds on every disk:

$$\begin{aligned} |t| &\leq \frac{13}{12} < 1.1, \\ |\lambda(t)| &\leq \frac{120}{\sqrt{3}(\sqrt{601} - 10)} < 4.774 < 5, \\ |z(t)| &\leq \frac{80}{\sqrt{601} - 10} + \frac{2}{5} < 5.912 < 6.2. \end{aligned}$$

For the two finer comparisons, it suffices to square the rational lower bounds $1.732^2 = 2.999824 < 3$ and $24.515^2 = 600.985225 < 601$, and then use

$$\begin{aligned} 4.774(1.732)(24.515 - 10) &= 120.01826452 > 120, \\ (5.912 - .4)(24.515 - 10) &= 80.00668 > 80. \end{aligned}$$

The image disk for $z = -B\zeta - ih$ has center $-B(c_j - ih)/\Delta_j - ih$ and radius $B\rho/\Delta_j$. Its lowest imaginary part is therefore $B(h - \rho)/\Delta_j - h$. Since $h > \rho$, this decreases with Δ_j , giving

$$\text{Im } z(t) \geq B \frac{h - 1/6}{c_j^2 + h^2 - (1/6)^2} - h \geq -\frac{1402}{17505} > -.081. \quad (\text{A.9})$$

Since $\sum_{l=1}^6 |P_l| = 10$ and $|t_l - t| \leq 13/12$ for $l \geq j$, the analytic continuation of (A.7) on this disk is bounded by

$$3 \max\{2\pi^2 \cdot 10 \cdot 13/12, 2\pi \cdot 10\} e^{100\pi/6} < 1000 e^{100\pi/6}.$$

Use $|a^l/l!| \leq e^{|a|}$ for the divided powers. Including all factors, the direct and transformed integrands are bounded, respectively, by

$$\begin{aligned} 1000 e^{200\pi/6 + 1.5\pi \cdot 1.1} &< 5.38 \cdot 10^{50}, \\ 5000 e^{100\pi/6 + 100\pi \cdot 0.81 + 1.5\pi \cdot 6.2} &< 1.51 \cdot 10^{50}. \end{aligned}$$

We may therefore use the common bound $M = 10^{53}$. The real interval has half-length $1/12$, half the disk radius. Cauchy's coefficient estimate bounds the tail of the Taylor polynomial of degree 255 by $M \sum_{k=256}^{\infty} 2^{-k} = M 2^{-255}$. Exactness, positivity and total mass one for the integral and quadrature rule give discrepancy at most

$$2^{-254} 10^{53} < 3.455 \cdot 10^{-24} < 10^{-22}. \quad (\text{A.10})$$

Atoms are evaluated separately and incur no quadrature error. $\square$

A.3 Interval evaluation

For a finite input p in the real finite-input domain of Lemma A.1 and a node row $m \in I_f$, form the two real numbers

$$\mathcal{S}_m(p) = \left(-\frac{X_{p,0}(m,1)}{Q_m}, \frac{D_m X_{p,0}(m,1) - X_{p,1}(m,1)}{Q_m} \right). \quad (\text{A.11})$$

With input $c_n = 1$ or $d_n = 1$ and all other data zero, these are the entries of the corresponding column of S at row node m .

Let $D = S_{J,J}$ and $U = S_{J,E}$.

For the Bernstein certificates, use exactly the triples (i, m, y) and orders ν specified in Lemma 4.1; in particular y is signed and no leftward interval at $m = 1$ is used for index 1. Express its power coefficients in terms of the moments as

$$T_r = \sigma_i y^{-\nu} (U_{\tilde{p}_i, r+\nu}(m, y) + X_{\tilde{p}_{3-i}, r+\nu}(m, y)). \quad (\text{A.12})$$

The signed scaling is exact. Definition (A.5) gives $U_{p,l}(m, y) = y^l U_{p,l}(m, 1)$ and the same identity for X . Therefore, for positive or negative y ,

$$T_r = \sigma_i y^r (U_{\tilde{p}_i, r+\nu}(m, 1) + X_{\tilde{p}_{3-i}, r+\nu}(m, 1)).$$

This is the identity used when the implementation caches unscaled moments at $y = 1$; it does not discard the orientation of a half-gap.

Equations (A.6), (A.7), (A.8), (A.11), and (A.12) already specify all sums in the finite certificates below. The following pseudocode records their algebraic organization. A measure is a list of pairs (t, g) in the order used by the supplied interval implementation: first the $6 \cdot 256$ quadrature pairs $(\tau_{jv}, o_v f_j(\tau_{jv})/6)$, with $j = 1, \dots, 6$ outermost and $v = 0, \dots, 255$ innermost, and then the seven atom pairs $(t_j, e_j C P_j)$ for $j = 0, \dots, 6$. The atom masses are zero for a basis input. The pseudocode is not a byte-for-byte execution trace. The supplied source fixes the detailed arithmetic and summation order for an actual interval run, including the factoring and caching described below.

```

Moments(measure, m, y, L):
    U[0..L] = 0; X[0..L] = 0
    for (t,g) in measure:
        lam = i/(b*(t+i*h))
        z = -B/(t+i*h)-i*h
        u = g*exp(i*pi*t*m)
        v = g*lam*exp(i*pi*z*m)
        for r = 0..L:
            U[r] += real(u); X[r] += real(v)
            u *= (i*pi*t*y)/(r+1)
            v *= (i*pi*z*y)/(r+1)
    return U,X

Pair(measure,m):
    U,X = Moments(measure,m,1,1)
    return -X[0]/Q[m], (D[m]*X[0]-X[1])/Q[m]

for basis column c[n] or d[n], n in I_f:
    measure = Measure(basis column; all other data zero)
    column = stack Pair(measure,m), m in J
    if n in J:
        also form rows m in E intersect [7,16]
        test column sums of W[eta] and the defect/product matrices

for i in {1,2}, m in I_f:
    c,d = Pair(Measure(full table input 3-i),m)
    test abs(table_c[i,m]-c)
    test abs(table_d[i,m]+indicator(i==1 and m==1)-d)

for (i,m,y,nu) in the prescribed half-gap cases:
    U,X = Moments for the two full table inputs, through L=30
    T[r] = sigma[i]*(U[i,r+nu]+X[3-i,r+nu])/y**nu
    for k = 0..28:
        Bk = T[0]; a = 1
        for r = 1..k:
            a *= (k+1-r)/(29-r)
            Bk += a*T[r]
        test the appropriate lower bound for Bk

```

Here **stack** means all first entries followed by all second entries. The recurrence for a equals $\binom{k}{r}/\binom{28}{r}$. **Measure** denotes this finite list and introduces no additional approximation. The fixed data at zero are used only for the two full table inputs. Reusing a measure means reusing its whole finite list.

For the accompanying certified evaluation, every real or complex quantity in these sums is enclosed using interval arithmetic. The pair calculation forms the moments at $y = 1$. For the Bernstein calculation, the implementation also forms and caches the unscaled moments at $y = 1$, then uses the signed y^r identity following (A.12). Each unscaled real moment enclosure is enlarged by $[-10^{-22}, 10^{-22}]$, as justified by Lemma A.1. Consequently the analytic-remainder contribution to the enclosure radius of a Bernstein coefficient is bounded by

$$\begin{aligned}
 2 \cdot 10^{-22} \sum_{r=0}^k \frac{\binom{k}{r}}{\binom{28}{r}} |y|^r &\leq 2 \cdot 10^{-22} \sum_{r=0}^{28} (3/2)^r \\
 &= 4 \cdot 10^{-22} ((3/2)^{29} - 1) < 5.2 \cdot 10^{-17}.
 \end{aligned}$$

The last inequality is an exact rational comparison. It is not correct to regard 10^{-22} as an unamplified per-Bernstein error. Arb rounding and dependency effects can further widen the balls; all of them are

propagated by the subsequent products, absolute values, column sums, and Bernstein combinations. An upper bound is accepted only when the complete enclosure lies strictly below the asserted rational bound, and a lower bound only when it lies strictly above. This procedure uses enclosures of the exact integrals, not a quadrature error inferred from observed numerical agreement.

The ball-arithmetic inclusion methodology is described in [6]. The supplied implementation uses Arb through python-flint at 192 bits of precision; the citation does not by itself certify this local environment, its call domains, or a run. The entry point `verification/verify_all.py` runs two verifiers in the same directory, `interval_certificate.py` and `interval_bounds.py`. Their output records, `finite.json` and `scalar.json`, and the aggregate record `summary.json` are placed in its `results/` subdirectory. The first verifier checks the exact bounds in (A.13), (A.14), (A.15), and (A.16), as well as all 2436 individual Bernstein inequalities in Lemma 4.1. It also asserts four finite brackets: the two defect brackets, the exterior bracket, and the bracket for $(2, 16, -1/2, 28)$. All its comparisons and extrema use complete enclosures.

The mathematical W argument in the next subsection uses only products and column sums. The supplied program also calls a certified inverse for $I - \eta D$ and tests its inverse and cross-product norms as additional checks; these calls are not needed for the W deduction.

A.4 Matrix and residual certificates

The following are strict upper bounds for the indicated column norms. The last three columns use only columns whose node lies in the stated range. They refer to the exact operator entries in (A.11).

η	$\ W_\eta\ $	$\ I - W_\eta(I - \eta D)\ $	$\ W_\eta U_{7:9}\ $	$\ W_\eta U_{12:16}\ $	$\ W_\eta U_{19:100}\ $
+1	30.84	.001	3.54	.44	.25
-1	1.83	.001	.73	.32	.17

(A.13)

The unmultiplied exterior block also satisfies

$$\|S_{E \cap [7,16], J}\| < .088. \quad (\text{A.14})$$

The certified enclosure calculation described in Section A.3 proves these inequalities. For example, the two defect norms are enclosed respectively in

$$\begin{aligned} & [.00025422624910652544, .00025422624910652546], \\ & [.00042288165102553355, .00042288165102553357], \end{aligned}$$

and the exterior block norm is enclosed in $[.08688748182289641591, .08688748182289641592]$. These intervals are stated with rational endpoints rounded outwards.

The three printed product ranges partition E , so $\|W_\eta U\|$ is the maximum of the three grouped product norms. The exact bounds in (A.13) and (A.14) give the primary matrix conclusion directly. Put $A_\eta = I - \eta D$. Since $\|I - W_\eta A_\eta\| < .001$, the finite square matrix $W_\eta A_\eta$ is invertible by a geometric series, hence so is A_η , and

$$A_\eta^{-1} = (W_\eta A_\eta)^{-1} W_\eta.$$

It follows from (A.13) that

$$\|A_\eta^{-1}\| < \frac{30.84}{1 - .001} < 32, \quad \|A_\eta^{-1} U\| < \frac{3.54}{1 - .001} < 3.7.$$

Equation (A.14) gives the remaining bound $< .09$ in Lemma 3.2.

For the two full tabulated inputs, including (A.1), the residual certificate is

$$\begin{aligned} \left| \tilde{c}_{i,m} + \frac{X_{\tilde{p}_{3-i},0}(m,1)}{Q_m} \right| &< 3 \cdot 10^{-10}, \\ \left| \tilde{d}_{i,m} + \mathbf{1}_{\{i=1, m=1\}} + \frac{X_{\tilde{p}_{3-i},1}(m,1) - D_m X_{\tilde{p}_{3-i},0}(m,1)}{Q_m} \right| &< 3 \cdot 10^{-10} \quad (i = 1, 2, m \in I_f). \end{aligned} \quad (\text{A.15})$$

The primary evaluation in Section A.3 proves (A.15) for these exact residuals. Since $3 \cdot 10^{-10} < 10^{-9}$, these inequalities, the preceding primary matrix conclusion, and (A.2) prove every finite assertion of Lemma 3.2.

A.5 Bernstein coefficient certificates

Let $B_{i,m,y,k}$ denote the sum B_k in (4.4). Each entry below is a strict lower bound for every such sum in its row, for all $0 \leq k \leq 28$ and both half-gap directions when applicable.

m	$i = 1$	$i = 2$
0	—	.762
1	.314	.191
3, 4	.028	.024
7, 9	.114	.112
12, 13, 15, 16	.0105	.0094
19, 21	.123	.113
24, 25, 27, 28	.0111	.0113
31, 33	.121	.126
36, 37, 39, 40	.0107	.0115

(A.16)

These bounds concern the exact derivatives of the tabulated functions, not derivatives rounded to satisfy interpolation identities. For example, the coefficient at $(i, m, y, k) = (2, 16, -1/2, 28)$ is enclosed in

$$[.00951875955680, .00951875955684].$$

The primary evaluation in Section A.3 proves the grouped bounds (A.16) for the exact derivatives of the tabulated functions. Each printed lower bound is stronger than its corresponding threshold in (4.4). Therefore these exact grouped bounds imply Lemma 4.1.

A.6 Scalar envelope and sign bounds

The remaining finite substitutions involve elementary real expressions. For the endpoint and density envelopes in (3.3), it suffices to test the finitely many endpoints and residues specified there. The endpoint formulas for $|\lambda|$, $|z|$, and $\text{Im } z$ use rational numbers and square roots. The density expressions additionally use sine and cosine at rational multiples of π . The endpoint tests are certified by the same elementary enclosures used above; their monotonicity and convexity extensions to whole intervals were proved in Section 3.

For the row envelope in (3.7), certified outward bounds are

row cut x	upper bound for the finite envelope sum
0	27.618723
4	.269094
16	.014122
100	$4.617010 \cdot 10^{-10}$

(A.17)

The atom-column analogue at cut 100 is less than $3.131186 \cdot 10^{-8}$. These bounds imply all the row-tail constants used in Section 3.

For the sign proof, the following outward bounds record the relevant substitutions. The expressions themselves, and the reason they bound whole intervals or tails, are given at the cited equations.

Expression	Certified bound
(4.13), index 1	$> .01206834$
(4.13), index 2	$> .01656468$
$\sum_{j=1}^6 M_j(\pi t_j)^2/6$	< 76.043
Right side of (4.14) before its final comparison	$< .005303$
(4.6) divided by $\nu!$, $(s_*, \nu) = (0, 0)$	$< .003189$
Same expression, $(s_*, \nu) = (1/2, 1)$	$< .003736$
Same expression, $(s_*, \nu) = (1/2, 2)$	$< .009146$
Same expression, $(s_*, \nu) = (2, 2)$	$< .001796$
$2.29 \sum_{j=1}^6 M_j/6 + .017 \sum_{j=0}^6 m_j^*$	< 72.669

In the fourth row the expression means the explicit sum with $.00005 \cdot 77$, not the derivative supremum to its left. For the five cases of (4.9), the corresponding bounds for $\mathcal{E}(m, Y, \nu)$ in (4.8) are

$$\begin{array}{c|c}
(m, Y, \nu) & \text{upper bound} \\
\hline
(0, 1/2, 0) & 1.578 \cdot 10^{-10} \\
(1, 1, 1) & 2.007 \cdot 10^{-5} \\
(1, 1, 2) & 5.735 \cdot 10^{-6} \\
(3, 1, 2) & 9.581 \cdot 10^{-11} \\
(4, 3/2, 2) & 6.133 \cdot 10^{-7}
\end{array} \tag{A.18}$$

The first two rational-part bounds can be obtained by exact fraction addition alone. All the other displayed scalar bounds follow by substitution and outward enclosure of the explicit exponential, trigonometric and rational expressions. The scalar verifier records 22 named strict comparisons: the four row substitutions, the atom tail, the nine sign-table rows, the five Taylor representatives, and the three disk-integrand and quadrature comparisons. It also checks the endpoint and residue envelopes, the coarser row-tail comparisons, the finite-inverse substitution giving a bound below 90, and the distance substitution using 91 and 2540. The derivation of these whole-space coefficients from the Schur estimates remains the analytic argument in Section 3. Its disk assertions use the coarser bounds 5 and 6.2 for $|\lambda|$ and $|z|$; the finer 4.774 and 5.912 bounds were proved analytically above and are not literal scalar-program assertions. At the six gap midpoints the program evaluates the product defining P and checks that each quotient is greater than .68; it does not assert the symbolic radical identities for those values. No sampling of an unknown function between interpolation nodes is involved. The logarithmic concavity argument and exact midpoint values in Lemma 4.2 supply the global lower bound for P .

B Alternative certification methods

The primary certificate does not use the methods in this appendix. We develop several ways to assess or reproduce its finite quantities. The first gives a fixed rational evaluation of every $N = 256$ moment and proves a uniform error theorem. A separate exact-rational program checks the first 6×6 block with an $N = 64$ rule. Rational-center Taylor integration gives a different, arbitrarily

refinable enclosure method, and the final proposition gives a local formula for the direct cardinal derivatives.

All terminating decimals denote exact rational numbers. The full $N = 256$ rational recipe and the Taylor method specify computations and their error bounds without asserting unreported finite values. The $N = 64$ program has the narrower scope stated with its proof.

B.1 Rational exponential primitives

The fixed rational procedures below use rational arithmetic alone, with a complex number represented by a pair of rationals. Put

$$\begin{aligned} p_* &= 3.141592653589793238462643383279502884, \\ b_* &= .866025403784438646763723170752936183, \end{aligned}$$

and retain $B = 4/3$ and $h = 2/5$ exactly. Both $|p_* - \pi|$ and $|b_* - b|$ are less than 10^{-36} . For example, the identity $\pi/4 = 4 \arctan(1/5) - \arctan(1/239)$ and alternating series through power 57 certify the first assertion; comparing $(b_* \pm 10^{-36})^2$ with $3/4$ certifies the second. Define

$$[w]_{40} = 10^{-40} \lfloor 10^{40} w \rfloor$$

for real w , and apply it componentwise to complex numbers. Define an approximation $e_*(u)$ to $e^{i\pi u}$ by the following finite recursion:

$$v_0 = \left[\sum_{q=0}^{32} \frac{(ip_* u / 1024)^q}{q!} \right]_{40}, \quad v_{a+1} = [v_a^2]_{40} \quad (0 \leq a < 10), \quad e_*(u) = v_{10}. \quad (\text{B.1})$$

Real and imaginary parts of $e_*(u)$ give the cosine and sine replacements when u is real.

Lemma B.1 (Rounded exponential). *For every $u \in \mathbb{Q}(i)$ satisfying $\Re(ip_* u) \leq 0$ and $|p_* u| \leq 1000$, the recursion (B.1) satisfies*

$$|e_*(u) - e^{ip_* u}| < 3.542 \cdot 10^{-34} < 10^{-32}.$$

Proof. The reduced argument has modulus less than one, so the Taylor remainder in the first step of (B.1) is less than $3/33!$. A componentwise floor contributes less than $2 \cdot 10^{-40}$ in complex absolute value. Each exact exponential subsequently squared has modulus at most one. Thus the recurrence

$$e_0 = 3/33! + 2 \cdot 10^{-40}, \quad e_{a+1} = 2e_a + e_a^2 + 2 \cdot 10^{-40}$$

bounds the accumulated error. Exact rational evaluation gives $e_{10} < 3.542 \cdot 10^{-34}$, as claimed. $\square$

Positive exponential upper bounds. For $x \geq 0$ and an integer $n \geq 0$ with $x < n + 2$, define

$$\mathcal{T}_n(x) = \frac{x^{n+1}}{(n+1)!} \frac{1}{1 - x/(n+2)}, \quad \mathcal{E}_n(x) = \sum_{k=0}^n \frac{x^k}{k!} + \mathcal{T}_n(x). \quad (\text{B.2})$$

After the first omitted term, the ratio of each term to its predecessor is at most $x/(n+2) < 1$, so $e^x \leq \mathcal{E}_n(x)$. These are rational upper bounds when x is rational. They will be used both in the fixed first-block check and in the arbitrary-refinement Taylor method.

B.2 A fixed rational evaluation

We now apply the fixed primitives to the full $N = 256$ rule, using rational coefficients and rational C, m, y in the finite domain of Lemma A.1. After defining the rounded sums, we prove their uniform approximation error. No matrix, residual, or Bernstein values are asserted by that error theorem.

In (A.6), replace each cosine $\cos(\pi u)$ by $\Re e_*(u)$, with $u = \alpha_v$ or $u = 2a\alpha_v$ as appropriate, and then apply $[\]_{40}$ to the final weight o_v only. Denote the results by $o_v^\#, \tau_{jv}^\#$. In the displayed coefficient list for P_l , replace b by b_* ; call the resulting numbers $P_l^\#$. Evaluate (A.7) by replacing π by p_* , P_l by $P_l^\#$, and every exponential by e_* , at the approximate point $t = \tau_{jv}^\#$. Call the resulting density $f_j^\#(t)$. For either these points or the exact atom points $t = t_j$, use

$$\lambda_* = [i/(b_*(t + ih))]_{40}, \quad z_* = [-B/(t + ih) - ih]_{40}, \quad (\text{B.3})$$

and set

$$E_l(t) = e_*(tm) \frac{(ip_*ty)^l}{l!}, \quad G_l(t) = \lambda_* e_*(z_*m) \frac{(ip_*z_*y)^l}{l!}. \quad (\text{B.4})$$

The rational moment sums are therefore

$$\begin{aligned} U_{p,l}^\# &= \Re \left\{ \frac{1}{6} \sum_{j=1}^6 \sum_{v=0}^{255} o_v^\# f_j^\#(\tau_{jv}^\#) E_l(\tau_{jv}^\#) + C \sum_{j=0}^6 e_j P_j^\# E_l(t_j) \right\}, \\ X_{p,l}^\# &= \Re \left\{ \frac{1}{6} \sum_{j=1}^6 \sum_{v=0}^{255} o_v^\# f_j^\#(\tau_{jv}^\#) G_l(\tau_{jv}^\#) + C \sum_{j=0}^6 e_j P_j^\# G_l(t_j) \right\}. \end{aligned} \quad (\text{B.5})$$

No additional floors are used. For $Q_n^\#, D_n^\#$ use the product and cotangent formulas defining Q_n, D_n , with the same sine and cosine replacements and residue $a = n \bmod 12 \in A$; cotangent means the ratio of the replaced cosine to the replaced sine.

For the finite pseudocode in Section A.3, the rational prescription uses the same measure order: the $6 \cdot 256$ quadrature pairs followed by the seven atom pairs. Put $\#$ on the prescribed quantities and make exactly the replacements above; no other rounding is performed.

Lemma B.2 (Rational moment and node errors). *Let $p = PR$ have a rational coefficient list supported in $\{0\} \cup I_f$, with $\|(c, d)\|_1 \leq 3$, and let $C \in \mathbb{Q}$ satisfy $|C| \leq 1$. Let $m, y \in \mathbb{Q}$ and $l \in \{0, \dots, 30\}$ satisfy $0 \leq m \leq 100$ and $|y| \leq 3/2$. The prescription in Section B.2 gives rational numbers $U_{p,l}^\#$ and $X_{p,l}^\#$ such that*

$$|U_{p,l}^\# - U_{p,l}| < 10^{-14}, \quad |X_{p,l}^\# - X_{p,l}| < 10^{-14}.$$

Separately, for every node $n \in 12\mathbb{Z} + A$, using its residue $a \in A$ in (2.8), the prescribed rational values satisfy

$$|Q_n^\# - Q_n| < 10^{-25}, \quad |D_n^\# - D_n| < 10^{-25}.$$

Proof. We verify the call hypotheses of Lemma B.1 before using its error bound, and then propagate the errors through the finite sums. The cosine calls defining the nodes and weights have real arguments $u = \alpha_v$ or $u = 2a\alpha_v$, with $|u| < 254$; hence $|p_*u| < 3.142 \cdot 254 < 1000$ and $\Re(ip_*u) = 0$. For real u , replacing p_* by π adds at most $|u|10^{-36}$. The telescoping coefficient sum in the weights is less than one, so

$$\begin{aligned} |\tau_{jv}^\# - \tau_{jv}| &< \frac{3.542 \cdot 10^{-34} + 10^{-36}}{12} < 9 \cdot 10^{-34}, \\ |o_v^\# - o_v| &< \frac{3.542 \cdot 10^{-34} + 254 \cdot 10^{-36}}{256} + 10^{-40} < \frac{10^{-29}}{256}. \end{aligned}$$

The exact nodes have a uniform distance from their piece endpoints:

$$\begin{aligned} \min\{\tau_{jv} - t_{j-1}, t_j - \tau_{jv}\} &\geq \frac{1 - \cos(\pi/512)}{12} = \frac{\sin^2(\pi/1024)}{6} \\ &\geq \frac{1}{6 \cdot 512^2} = \frac{1}{1572864} > 9 \cdot 10^{-34}. \end{aligned}$$

Here $\sin x \geq 2x/\pi$ on $[0, \pi/2]$ follows from concavity. It follows that every real rounded node $\tau_{jv}^\#$ remains strictly inside (t_{j-1}, t_j) . The density is therefore evaluated on its stated piece, and all rounded nodes and exact atoms lie in $[0, 1]$. The remaining real exponential calls in the density and direct factors now have $|u| \leq 100$; the residue arguments for Q_n, D_n have $|u| < 1$. They too satisfy the call hypotheses.

For real $t \in [0, 1]$, the exact identities

$$\begin{aligned} |\lambda(t)| &\leq \frac{5}{\sqrt{3}} < 3, \\ |z(t)|^2 &= h^2 + \frac{B^2 - 2Bh^2}{t^2 + h^2} \leq \left(\frac{44}{15}\right)^2, \\ \text{Im } z(t) &= h \left(\frac{B}{t^2 + h^2} - 1 \right) \geq \frac{26}{435} > .0597 \end{aligned}$$

use $B^2 - 2Bh^2 = 304/225 > 0$. They apply in particular at every rounded node before the floor in (B.3). That floor lowers the imaginary part by less than 10^{-40} and changes the complex value by less than $2 \cdot 10^{-40}$. Thus

$$\text{Im } z_* \geq \frac{26}{435} - 10^{-40} > 0, \quad |z_*| < \frac{44}{15} + 2 \cdot 10^{-40} < 3.$$

For the exact transformed arguments, $100\pi|z(t)| < 922$; for the rounded ones, $|p_* z_* m| < 100(3.142)(3) < 1000$ and $\Re(ip_* z_* m) = -p_* m \text{Im } z_* \leq 0$. The exponential estimate already proved therefore applies to every transformed call as well.

For a quadrature node write $t = \tau_{jv}$ and $t^\# = \tau_{jv}^\#$; at an atom put $t = t^\# = t_j$. Along the intervening real segment, $|t + ih| \geq h$ and $b, b_* > .86$. The derivative bounds for the reciprocal factors and the change of b give

$$\begin{aligned} |\lambda_* - \lambda(t)| &< 8(9 \cdot 10^{-34}) + 4 \cdot 10^{-36} + 2 \cdot 10^{-40} < 10^{-29}, \\ |z_* - z(t)| &< \frac{25}{3}(9 \cdot 10^{-34}) + 2 \cdot 10^{-40} < 10^{-29}. \end{aligned}$$

Indeed $1/(b_* h^2) < 8$, $B/h^2 = 25/3$, and the change of $1/b$ contributes less than $10^{-36}/(.86^2 h) < 4 \cdot 10^{-36}$. At atoms the node-error terms vanish. Both transformed exponential arguments lie in the closed left half-plane after multiplication by i . The integral first-difference formula there gives, for example,

$$|e_*(z_* m) - e^{i\pi z(t)m}| < 10^{-32} + 100(3.142 \cdot 10^{-29} + 3 \cdot 10^{-36}) < 4 \cdot 10^{-27}.$$

The direct exponential-factor error is smaller. The arguments of the divided powers differ by less than $1.5(3.142 \cdot 10^{-29} + 3 \cdot 10^{-36}) < 5 \cdot 10^{-29}$. The divided powers and their derivatives are bounded along the connecting segment by $e^{1.5 \cdot 3.142 \cdot 3} < 1.5 \cdot 10^6$; this also gives the stated bound $e^{1.5\pi \cdot 3} < 1.5 \cdot 10^6$. One rational check of this exponential bound uses $1.5(3.142)(3)/16 < .884$ and

$$\begin{aligned} \sum_{q=0}^8 \frac{.884^q}{q!} + \frac{.884^9}{9!} \frac{1}{1 - .884/10} &< 2.421, \\ 2.421^{16} &< 1.5 \cdot 10^6. \end{aligned}$$

The first expression majorizes the exponential series because every successive ratio in its tail is at most $.884/10$. The divided-power errors are therefore less than $7.5 \cdot 10^{-23} < 1.5 \cdot 10^{-22}$. Using $|\lambda| < 3$, modulus at most one for the exact exponentials, and modulus less than two for their approximations, the transformed-factor error is at most

$$2(1.5 \cdot 10^6)10^{-29} + 3(1.5 \cdot 10^6)(4 \cdot 10^{-27}) + 3(1.5 \cdot 10^{-22}) < 3 \cdot 10^{-20}.$$

The same bound applies to the direct factor.

For completeness, the density estimate can be obtained without relying on cancellation. Its real exponential factors, including evaluation at a rounded node, have error less than $3.542 \cdot 10^{-34} + 100(3.142(9 \cdot 10^{-34}) + 10^{-36}) < 3 \cdot 10^{-31}$. Let A_n, B_n denote the inner sums in (A.7) with and without the factor $t_l - t$, respectively, and put $\#$ on their replacements. On the common piece, $|t_l - t|, |t_l - t^\#| \leq 1$ for $l \geq j$. The displayed list for P_l gives $\sum_{l=1}^6 |P_l^\# - P_l| < 7 \cdot 10^{-36}$ and $\sum_{l=1}^6 |P_l^\#| < 11$. Hence $|A_n^\#|, |B_n^\#| < 22$ and

$$\begin{aligned} |A_n^\# - A_n|, |B_n^\# - B_n| &< 22(9 \cdot 10^{-34}) + 2(7 \cdot 10^{-36}) \\ &+ 10(3 \cdot 10^{-31}) < 4 \cdot 10^{-30}. \end{aligned}$$

The bracket in (A.7) has approximate magnitude at most $70(|c_n| + |d_n|)$ and error less than $1.3 \cdot 10^{-29}(|c_n| + |d_n|)$. Expanding the outer product once and using list norm at most three gives the pointwise bound

$$\begin{aligned} |f_j^\#(t^\#) - f_j(t)| &< 2(10^{-36})(3)(2)(70) \\ &+ 2(3.142)(3)(70(3 \cdot 10^{-31}) + 1.3 \cdot 10^{-29}) \\ &< 7 \cdot 10^{-28} < 10^{-22}. \end{aligned}$$

On the real segment $|f_j| < 1000$, and each atom has magnitude at most six and error less than 10^{-28} . The latter error also follows directly from the displayed P_l list. The exact transformed factor has magnitude below $3 \cdot 1.5 \cdot 10^6$, so every approximate factor has magnitude below $4.6 \cdot 10^6$. The exact quadrature weights have total mass one, and the sum of the absolute weight errors over all six pieces is less than 10^{-29} . Separating weight, density, factor, and atom errors, and then adding the analytic quadrature error, bounds either moment error by

$$\begin{aligned} &10^{-29}(1001)(4.6 \cdot 10^6) + 10^{-22}(4.6 \cdot 10^6) + 1000(3 \cdot 10^{-20}) \\ &+ 7(10^{-28}(4.6 \cdot 10^6) + 6(3 \cdot 10^{-20})) + 10^{-22} < 5 \cdot 10^{-16} < 10^{-14}. \end{aligned}$$

Finally, for the residue arguments the sine and cosine replacement errors are less than $3.542 \cdot 10^{-34} + 10^{-36} < 4 \cdot 10^{-34}$. Every nonzero exact sine has magnitude at least $\sin(\pi/12) > 1/4$, and its replacement has magnitude greater than $1/5$. Each squared factor $(2 \sin)^2$ in Q_n and its replacement is less than five in magnitude, and their difference is less than $5 \cdot 10^{-33}$. Telescoping the five factors and allowing for $(p_*/6)^2 - (\pi/6)^2$ gives a Q_n error less than $2 \cdot 10^{-29} < 10^{-25}$. For a cotangent, division by the two sine lower bounds gives an error less than $5(4 \cdot 10^{-34}) + 20(4 \cdot 10^{-34}) = 10^{-32}$. Summing five terms and allowing for the prefactor gives a D_n error less than $6 \cdot 10^{-32} < 10^{-25}$, as required. $\square$

B.3 Propagation to finite certificates

We record stability estimates before applying any numerical margin. The first controls the two operator coordinates under errors in the moments and node constants. The second controls Bernstein sums from errors in the signed- y moments.

To propagate a moment radius, abbreviate $X_\ell = X_{p,\ell}(m, 1)$ and write $\mathcal{S}_{m,c}, \mathcal{S}_{m,d}$ for the two components of (A.11), with hats for their rational approximations. Suppose $|X_0|, |X_1| \leq H$, $|D_m| \leq d$, and rational approximations have errors at most $\epsilon, \epsilon, \eta_Q, \eta_D$ in X_0, X_1, Q_m, D_m , respectively. If both exact and approximate Q_m exceed $q > 0$, expanding the numerator and then the reciprocal in (A.11) gives

$$\max\{|\widehat{\mathcal{S}}_{m,c} - \mathcal{S}_{m,c}|, |\widehat{\mathcal{S}}_{m,d} - \mathcal{S}_{m,d}|\} \leq \frac{(d+1+\eta_D)\epsilon + H\eta_D}{q} + \frac{(d+1)H\eta_Q}{q^2}. \quad (\text{B.6})$$

For example, the perturbed second numerator differs by at most $(d+\eta_D)\epsilon + H\eta_D + \epsilon$, and the exact numerator is bounded by $(d+1)H$; this proves the inequality, which also dominates the first entry's error. The Taylor budgets and the separated and common-kernel first-block estimates below use $q = 7/4$ and $d = 2.12$. The separate-rounding comparison uses $q = 7/4$ and $d = 4.3$. The rational lower bound $Q_m > 197192/112500$ proved after (2.9) exceeds $7/4$ by $1268/450000$, so $\eta_Q \leq 10^{-25}$ also ensures that the approximate Q_m exceeds q .

Analytic pair-error allowance. For an input p in the rational finite-input domain of Lemma B.2 and a row $m \in I_f$, define the rational version $\mathcal{S}_m^\#$ by using the quantities with superscript $\#$ throughout (A.11). We record the division estimate explicitly for both basis and full-table inputs. By (3.3) and the atom masses there, the total variation of the folded measure for any input in the real finite-input domain of Lemma A.1 is at most

$$3 \sum_{j=1}^6 \frac{M_j}{6} + \sum_{j=0}^6 m_j^* = 119.642 < 133.$$

On the real segment, $|\lambda|, |z| < 3$ and $|e^{i\pi z m}| \leq 1$ for $m \geq 0$. Thus, using $\pi < 22/7$,

$$|X_{p,0}(m, 1)| < 133(3) < 400, \quad |X_{p,1}(m, 1)| < 133(3)(3)(22/7) = 3762 < 4000.$$

For this calculation abbreviate $X_j = X_{p,j}(m, 1)$, $Q = Q_m$, and $D = D_m$. Write $\varepsilon = 10^{-14}$ and $\delta = 10^{-25}$. By Lemma B.2, the errors in X_0, X_1 are less than ε and those in Q, D are less than δ . Equation (2.9) gives $Q > 1.75$, $|D| < 2.12$, and hence $Q^\# > 1.7$. For $\mathcal{A} = DX_0 - X_1$ and $\mathcal{A}^\# = D^\#X_0^\# - X_1^\#$,

$$|\mathcal{A}| < 4848, \quad |\mathcal{A}^\# - \mathcal{A}| < 2.12\varepsilon + (400 + \varepsilon)\delta + \varepsilon < 3.121 \cdot 10^{-14}.$$

The two quotient differences are consequently bounded by

$$\begin{aligned} \left| \frac{X_0^\#}{Q^\#} - \frac{X_0}{Q} \right| &< \frac{10^{-14}}{1.7} + \frac{400 \cdot 10^{-25}}{1.75(1.7)} < 6 \cdot 10^{-15}, \\ \left| \frac{\mathcal{A}^\#}{Q^\#} - \frac{\mathcal{A}}{Q} \right| &< \frac{3.121 \cdot 10^{-14}}{1.7} + \frac{4848 \cdot 10^{-25}}{1.75(1.7)} < 1.84 \cdot 10^{-14} < 2 \cdot 10^{-14}. \end{aligned}$$

In particular, the uniform error in each entry of $\mathcal{S}_m^\#$ is strictly less than 10^{-12} ; we retain this looser allowance below.

B.3.1 Bernstein multipliers

For every signed half-gap in Lemma 4.1, $y \neq 0$ and $|y|^{-\nu} \leq 4$. Directly from (A.5),

$$y^{-\nu} (U_{p,r+\nu}(m, y) + X_{q,r+\nu}(m, y)) = y^r (U_{p,r+\nu}(m, 1) + X_{q,r+\nu}(m, 1)).$$

This identity also holds for negative y and gives exactly the power coefficients of (A.12); it does not replace their derivatives by interpolated jets. Two moment errors less than 10^{-14} give a T_r error less than $8 \cdot 10^{-14}$. For fixed $0 \leq k \leq 28$, the rational recurrence $a_0 = 1$ and $a_r = a_{r-1}(k+1-r)/(29-r)$ satisfies

$$a_r = \prod_{j=1}^r \frac{k+1-j}{29-j} = \frac{k!}{(k-r)!} \frac{(28-r)!}{28!} = \frac{\binom{k}{r}}{\binom{28}{r}} \quad (0 \leq r \leq k). \quad (\text{B.7})$$

Each factor is in $[0, 1]$. There are $\sum_{k=0}^{28} k = 28 \cdot 29/2 = 406$ nonconstant multipliers, in addition to $a_0 = 1$ for each k . Every Bernstein sum has at most 29 terms, so its error is strictly less than $29(8 \cdot 10^{-14}) = 2.32 \cdot 10^{-12}$.

B.4 An independent rational check of the first block

The optional program `verification/check_sjj_rational.py` checks the 6×6 block on $J = \{1, 3, 4\}$ by exact rational arithmetic. It uses the cosine rule (A.6) with $N = 64$ and the sum over $1 \leq a \leq 31$, and otherwise the rounded exponential, kernels and basis densities specified above. There are no atom terms for these basis inputs. The following error bound makes this smaller check independent of the full Arb calculation.

Lemma B.3. *For each basis input $c_n = 1$ or $d_n = 1$, $n \in J$, with all other data zero, and each output node $m \in J$, the rational program computes both entries of $\mathcal{S}_m(p)$ with absolute error less than 10^{-12} .*

Proof. We separate quadrature error from rounding in the finite sum. Use the same complex disks of radius $1/6$ as in Lemma A.1. The basis density on any disk is bounded by

$$D_0 = 2\pi^2 \cdot 10 \cdot \frac{13}{12} e^{4\pi/6}.$$

This also bounds a simple-pole basis density. The bounds $|\lambda| < 5$, $|z| < 6.2$ and $\text{Im } z > -.081$ proved there show that the transformed value and first-derivative integrands are bounded respectively by

$$M_0 = 5D_0 e^{4\pi \cdot 0.081} < 24028, \quad M_1 = 6.2\pi M_0 < 468015.$$

Use the rational upper bound $e^x \leq \mathcal{E}_n(x)$ from (B.2). Put $\bar{\pi} = p_* + 10^{-36} > \pi$. Here $x = 743\pi/750$ lies in $[0, 4)$, so the formula applies with $n = 40$. Using $M_0 = (325/3)\pi^2 e^{743\pi/750}$, the exact rational comparisons

$$\frac{325}{3} \bar{\pi}^2 \mathcal{E}_{40}(743\pi/750) < 24028, \quad \frac{31}{5} \bar{\pi} \frac{325}{3} \bar{\pi}^2 \mathcal{E}_{40}(743\pi/750) < 468015$$

prove the displayed bounds for M_0, M_1 . Each comparison is a fraction inequality after clearing positive denominators; no rounded decimal evaluation of an exponential is used. Discrete cosine orthogonality now gives exactness through degree 63. The weights sum to one and each is at least $1/(64 \cdot 63) > 0$ by the same telescoping identity. Cauchy's estimate on the half-radius real segments

bounds the degree-63 Taylor tails by $M_i 2^{-63}$. The integral and quadrature have total mass one, after combining all six pieces. Thus their discrepancies in the two moments are at most

$$\epsilon_0 = 2^{-62} M_0 < 5.211 \cdot 10^{-15}, \quad \epsilon_1 = 2^{-62} M_1 < 1.015 \cdot 10^{-13}.$$

Using $Q_m > 1.75$ and $|D_m| < 2.12$, the quadrature error in either entry of (A.11) is consequently less than

$$\frac{2.12\epsilon_0 + \epsilon_1}{1.75} < 6.432 \cdot 10^{-14}. \quad (\text{B.8})$$

For completeness, two changes in the rational implementation require care in reusing the rounding estimates. First, for a real argument u the program evaluates $e_*(r)$ with $r = u \bmod 2 \in [0, 2)$. Periodicity is used for the exact target: $e^{i\pi r} = e^{i\pi u}$. The approximation p_* is not treated as giving an exactly periodic exponential. Since r is real and $|p_* r| < 2(3.142) < 1000$, Lemma B.1, followed by $|e^{ip_* r} - e^{i\pi r}| \leq 2 \cdot 10^{-36}$, gives an absolute error less than $2 \cdot 10^{-32}$ for each such replacement. When u contains an approximate quadrature node, first compare at that approximate argument and then use the first-difference bound for the exact exponential. This avoids any assumption of continuity of the modulo operation.

Second, with $N = 64$, the sum of the absolute cosine coefficients in one weight is less than one. Including the final floor, the weight error is therefore at most

$$\frac{2 \cdot 10^{-32}}{N} + 10^{-40} < \frac{10^{-29}}{N}.$$

Summing the stronger first expression, rather than the relaxed bound on the right, gives the total weight error

$$N \left(\frac{2 \cdot 10^{-32}}{N} + 10^{-40} \right) = 2 \cdot 10^{-32} + 64 \cdot 10^{-40} < 3 \cdot 10^{-32}. \quad (\text{B.9})$$

The node error is less than $2 \cdot 10^{-32}/12 < 2 \cdot 10^{-33}$. The exact nodes are at distance at least $1/98304$ from the endpoints of their own interval $[(j-1)/6, j/6]$, since $1 - \cos(\pi/128) \geq 2/128^2$. Thus the approximate nodes remain in their own intervals, in particular in $[0, 1]$. Their transformed frequencies have positive imaginary part even after the floor, because $\text{Im } z \geq 26/435$ there. The real-segment estimates in the proof of Lemma B.2 also give $|z_*| < 3$. Thus every transformed call has its argument in the closed left half-plane and satisfies $|p_* z_* m| < 3.142(3)(4) < 1000$.

Here is a direct propagation bound for these smaller inputs. The real-segment estimates are $|\lambda| < 3$, $|z| < 3$, and basis-density modulus less than 214. For one piece put

$$A_n = \sum_{l=j}^6 P_l e^{i\pi t_l n}, \quad B_n = \sum_{l=j}^6 t_l P_l e^{i\pi t_l n}.$$

Their moduli are at most ten. Replacing b and the real exponentials changes each sum by less than $3 \cdot 10^{-31}$; including the node movement changes $B_n - tA_n$ by less than $7 \cdot 10^{-31}$. The phase $e^{-i\pi t n}$ changes by less than $5 \cdot 10^{-32}$, since $n \leq 4$. Substitution in (2.14) therefore gives a density error less than $3 \cdot 10^{-29}$. The derivatives of the rational kernels, using $b, b_* > 4/5$ and $|t + ih| \geq 2/5$, give errors less than $2 \cdot 10^{-32}$ in both λ and z , including their floors. The first-difference formula for exponentials in the closed left half-plane, with $m \leq 4$, then bounds the errors in

$$g_0(t) = \lambda(t) e^{i\pi m z(t)}, \quad g_1(t) = g_0(t) i\pi z(t)$$

by $2 \cdot 10^{-30}$ and $3 \cdot 10^{-29}$, respectively. The exact factors have moduli below 3 and 36; their approximations have moduli below 4 and 37. Product differences bound each derivative-integrand error by

$$3 \cdot 10^{-29} \cdot 37 + 214 \cdot 3 \cdot 10^{-29} < 8 \cdot 10^{-27};$$

the value-integrand bound is smaller. By (B.9), the total absolute weight error is below $3 \cdot 10^{-32}$, and the approximate integrands have modulus below 10^4 . Hence each rounded moment differs from its exact finite quadrature moment by less than 10^{-26} .

Finally, for residues 1, 3, 4, the exact polynomial forms are

$$\begin{aligned} Q_1 = Q_3 &= \frac{4\pi^2}{3}(1-b), & Q_4 &= \frac{\pi^2}{3}, \\ D_1 &= \frac{\pi(3+2b)}{18}, & D_3 &= -\frac{\pi(3+2b)}{18}, & D_4 &= \frac{7\pi b}{9}. \end{aligned}$$

The program replaces π, b by p_*, b_* in these expressions. Using $|p_*^2 - \pi^2| < 6.3 \cdot 10^{-36}$, $\pi^2 < 10$, and $b, b_* < 1$ gives

$$\begin{aligned} |Q_m^\# - Q_m| &< \max\{(4/3)(6.3+10), 6.3/3\}10^{-36} < 3 \cdot 10^{-35}, \\ |D_m^\# - D_m| &< \max\{(5+44/7)/18, 29/9\}10^{-36} < 4 \cdot 10^{-36}. \end{aligned}$$

The exact rational lower bound used after (B.6) therefore shows that both denominators $Q_m, Q_m^\#$ exceed $7/4$. The moments have modulus below 10^4 , so the additional division and D_m errors are less than $4 \cdot 10^{-31}$. Together with (B.8), these bounds give an error less than $7 \cdot 10^{-14} < 10^{-12}$ in either matrix entry, as asserted. $\square$

If every entry of this 6×6 matrix has error below $\varepsilon = 10^{-12}$, its induced column-norm error is below 6ε . The program accordingly enlarges the computed defect norm by $\|W_\eta\|_1 6\varepsilon$, then verifies that the result is below .003 and that the resulting inverse bound is below 32. This is the same perturbation-of-identity argument as in Section A.4. It checks both small inverse certificates without asserting that the full $N = 256$ rational prescription has been executed.

B.5 Comparison of first-block majorants

The preceding $N = 64$ proof separates the value and derivative integrands. Two coarser arguments instead use one majorant for both. They concern the same basis inputs $n \in J$, row nodes $m \in J$, $y = 1$, and orders zero and one, with no atoms. The table compares their disk and node bounds with the resulting entry errors:

route	disk majorant	$ D_m $	entry error
separated	$M_0 < 24028, M_1 < 468015$	< 2.12	$< 7 \cdot 10^{-14}$
common kernel	$M = 10^6$	< 2.12	$< 4.1 \cdot 10^{-13}$
separate rounding	$M = 750000$	< 4.3	$< 5.23 \cdot 10^{-13}$

The separated route uses the explicit density bound D_0 from Section B.4. The two common routes use density and exponential bounds (1800, 2.8) and (2500, 3), respectively.

B.5.1 A common kernel majorant

The rational comparisons

$$\mathcal{E}_{32}(44/21) < 42/5, \quad \mathcal{E}_{32}(891/875) < 14/5 \tag{B.10}$$

follow by clearing positive denominators in (B.2); both arguments are nonnegative and less than 34. Since $4\pi/6 < 44/21$ and $4\pi(.081) < 891/875$, the two basis densities on the complex disks satisfy

$$\begin{aligned} |u_n(t)| &< 2(22/7)^2(65/6)(42/5) = 12584/7 < 1800, \\ |v_n(t)| &< 2(22/7)10(42/5) = 3696/7 < 1800. \end{aligned}$$

The bounds $|\lambda| < 5$, $|e^{i\pi zm}| < 2.8$, and $\pi|z| < 6.2\pi < 20$ therefore bound both transformed integrands by $1800(5)(2.8)(20) = 504000 < 10^6$. The positive degree-63 rule and the half-radius Cauchy estimate give, for each moment,

$$|X_\ell - X_{\ell,\text{quad}}| \leq 10^6 2^{-62} < 2.17 \cdot 10^{-13}. \quad (\text{B.11})$$

The factor two here compares the integral and quadrature averages; a direct Taylor average has only one tail, as in Lemma B.4.

The rounding proof in Section B.4 applies to these same inputs and gives finite moment-rounding error below 10^{-26} , hence below 10^{-14} . It also gives $|X_0|, |X_1| < 10^4$, $|\delta Q| < 3 \cdot 10^{-35}$, $|\delta D| < 4 \cdot 10^{-36}$, and both denominators above $7/4$. Thus the total moment allowance is $2.27 \cdot 10^{-13}$. In (B.6), the additional Q, D contribution is less than

$$\frac{10^4(4 \cdot 10^{-36}) + (4 \cdot 10^{-36})(2.27 \cdot 10^{-13})}{7/4} + \frac{3.12 \cdot 10^4(3 \cdot 10^{-35})}{(7/4)^2} < 4 \cdot 10^{-31}.$$

Consequently

$$\frac{3.12(2.27 \cdot 10^{-13})}{1.75} + 4 \cdot 10^{-31} < 4.1 \cdot 10^{-13} < 10^{-12}. \quad (\text{B.12})$$

B.5.2 A separate rounding majorant

The weaker bounds $e^{4\pi/6} < 9$ and $e^{4\pi(.081)} < 3$ also follow from (B.10). They bound either basis density by 2500; for the larger density, $2(22/7)^2(65/6)9 < 2500$. With the kernel bound 5 and derivative factor $6.2\pi < 20$, this gives the distinct common majorant $M = 2500(5)(3)(20) = 750000$. Its moment quadrature radius is

$$\varepsilon_q = 750000 2^{-62} < 1.626303258728257 \cdot 10^{-13}. \quad (\text{B.13})$$

Using $|D_m| < 4.3$ and $Q_m > 7/4$, the pair quadrature error is less than $(5.3/(7/4))\varepsilon_q < 4.926 \cdot 10^{-13}$. The same finite rounding allowance below 10^{-14} applies because the schedule and inputs have not changed. Substitution of the same polynomial Q, D errors in (B.6), now with $d = 4.3$, bounds their additional contribution by $6 \cdot 10^{-31}$. Therefore

$$\frac{5.3}{7/4} (750000 2^{-62} + 10^{-14}) + 6 \cdot 10^{-31} < 5.23 \cdot 10^{-13} < 10^{-12}. \quad (\text{B.14})$$

All three routes support the program's 10^{-12} entry radius. They do not extend its coverage: the public rational algorithm still checks only the 36 first-block entries and the two inverse certificates, not the residual or Bernstein tables.

B.6 Rational-center Taylor integration

For a finite input in the real moment domain of Lemma A.1, let $F_{j,\ell}^U$ and $F_{j,\ell}^X$ be the density (A.7) multiplied by the respective factors in (A.8). The input is supported on nodes at most 100, has

coefficient-list norm at most 3 and $|C| \leq 1$, and the parameters satisfy $0 \leq m \leq 100$, $|y| \leq 3/2$, and $\ell \in \{0, \dots, 30\}$. The coefficients, C, m, y are rational when a rational enclosure is requested. The six centers and the two radii are

$$c_j = \frac{2j-1}{12}, \quad R = \frac{1}{6}, \quad a = \frac{1}{12} \quad (1 \leq j \leq 6). \quad (\text{B.15})$$

Each $F_{j,\ell}^U$ and $F_{j,\ell}^X$ is analytic on a neighborhood of $|t - c_j| \leq R$: the input list is finite and the only rational-kernel pole, $-ih$, is outside that disk. This analytic assertion is made for finite inputs. An arbitrary ℓ^1 list need not give a complex-frequency density analytic on these disks.

Lemma B.4 (Taylor averages with rational coefficient radii). *Suppose $F(c_j + x) = \sum_{k \geq 0} A_k x^k$ is analytic on a neighborhood of $|x| \leq R$ and $|F| \leq M$ there. Let $D \geq 0$ be an integer. For $0 \leq k \leq D$, suppose $\hat{A}_k \in \mathbb{Q}(i)$ and $e_k \in \mathbb{Q}_{\geq 0}$ satisfy $|A_k - \hat{A}_k| \leq e_k$. Then*

$$\left| 6 \int_{t_{j-1}}^{t_j} F(t) dt - \sum_{2r \leq D} \hat{A}_{2r} \frac{a^{2r}}{2r+1} \right| \leq M 2^{-D} + \sum_{2r \leq D} e_{2r} \frac{a^{2r}}{2r+1}. \quad (\text{B.16})$$

If the final sum of radii is at most δ for each of the six pieces, their combined integral has error at most $M 2^{-D} + \delta$. Discrete atoms are not included in this estimate and are to be enclosed separately.

Proof. Since $2a = 1/6$, the expression $6 \int F$ is the interval average. The average of x^k on $[-a, a]$ is zero for odd k and is $a^k/(k+1)$ for even k . Thus the degree- D Taylor polynomial has the exact average

$$\sum_{2r \leq D} A_{2r} \frac{a^{2r}}{2r+1}. \quad (\text{B.17})$$

Cauchy's estimate gives $|A_k| \leq M R^{-k}$. Bounding the average of the absolute tail by its supremum on $[-a, a]$ gives the conservative bound

$$M \sum_{k > D} (a/R)^k = M \sum_{k > D} 2^{-k} = M 2^{-D}.$$

Replacing the even coefficients adds the displayed weighted sum of their radii. Finally, each of the six integrals is $1/6$ times its average, so the six factors $1/6$ sum to one. Taking a real part cannot increase the error. Atoms have no integral tail of this kind. $\square$

Here are explicit rational operations for obtaining the coefficient enclosures in this lemma. Write $\mathcal{B}(a, r) = \{w \in \mathbb{C} : |w - a| \leq r\}$, where $a \in \mathbb{Q}(i)$ and $r \in \mathbb{Q}_{\geq 0}$, and put $\langle a \rangle_1 = |\Re a| + |\Im a|$. The following enclosure operations use rational centers and radii:

$$\begin{aligned} \mathcal{B}(a, r) + \mathcal{B}(b, s) &\subseteq \mathcal{B}(a + b, r + s), \\ \mathcal{B}(a, r) \mathcal{B}(b, s) &\subseteq \mathcal{B}(ab, \langle a \rangle_1 s + \langle b \rangle_1 r + rs), \\ \mathcal{B}(a, r)^{-1} &\subseteq \mathcal{B}(a^{-1}, r/\ell_0^2) \quad \text{if } 0 < \ell_0 \leq |a| - r. \end{aligned} \quad (\text{B.18})$$

In the last line $a^{-1} \in \mathbb{Q}(i)$, and the condition on a rational ℓ_0 can be checked by squaring rational quantities. The bound follows from $|w^{-1} - a^{-1}| = |w - a|/(|w||a|)$. These operations allow repeated use of the same enclosure; possible dependence between its occurrences does not invalidate inclusion.

Constant exponentials also admit arbitrarily small rational radii. Use the rational bounds $\mathcal{T}_n, \mathcal{E}_n$ from (B.2). If $w \in \mathcal{B}(w_0, r)$, choose rational $q \geq |w_0|$, for example $q = \langle w_0 \rangle_1$, and nonnegative integers n, s with $q < n + 2$ and $q + r < s + 2$. Then

$$e^w \in \mathcal{B} \left(\sum_{k=0}^n \frac{w_0^k}{k!}, \mathcal{T}_n(q) + r\mathcal{E}_s(q+r) \right). \quad (\text{B.19})$$

Indeed, the first radius bounds the complex Taylor tail at w_0 . The integral first-difference formula for e^w bounds the change from w_0 to w by re^{q+r} , giving the second radius. Both preconditions in (B.2) are explicit; no estimate of a positive exponential by a truncated sum alone is being used. For bounded q , $\mathcal{T}_n(q) \rightarrow 0$ as $n \rightarrow \infty$.

For clarity, the coefficient recurrences needed for the present integrands are as follows. If $a(x) = \sum a_k x^k$ and $b(x) = \sum b_k x^k$, multiplication is convolution,

$$(ab)_k = \sum_{v=0}^k a_v b_{k-v}. \quad (\text{B.20})$$

If $a_v \in \mathcal{B}(\hat{a}_v, e_v)$ and $b_v \in \mathcal{B}(\hat{b}_v, f_v)$, a rational radius for the center $\sum_{v=0}^k \hat{a}_v \hat{b}_{k-v}$ is explicitly

$$\sum_{v=0}^k \left(\langle \hat{a}_v \rangle_1 f_{k-v} + \langle \hat{b}_{k-v} \rangle_1 e_v + e_v f_{k-v} \right).$$

For a reciprocal series $q(x) = a(x)^{-1}$ with $a_0 \neq 0$, and for an exponential series $E(x) = e^{g(x)}$, the recurrences are

$$\begin{aligned} q_0 &= a_0^{-1}, & q_k &= -q_0 \sum_{v=1}^k a_v q_{k-v} \quad (k \geq 1), \\ E_0 &= e^{g_0}, & E_k &= \frac{1}{k} \sum_{v=1}^k v g_v E_{k-v} \quad (k \geq 1). \end{aligned} \quad (\text{B.21})$$

The second line follows by comparing coefficients in $E' = g'E$. Use (B.19) for E_0 and (B.18) in these recurrences. Along with the displayed convolution radius, this specifies a rational radius for every coefficient, not merely a formal power series.

More particularly, $d_j = c_j + ih \in \mathbb{Q}(i)$ and

$$r_0 = d_j^{-1}, \quad r_{k+1} = -r_k/d_j, \quad \lambda_k = ir_k/b, \quad z_0 = -Br_0 - ih, \quad z_k = -Br_k \quad (k \geq 1) \quad (\text{B.22})$$

are the coefficients of $(t + ih)^{-1}$, $\lambda(t)$, and $z(t)$ at $t = c_j + x$. The r_k, z_k are rational complex numbers; an enclosure for b^{-1} gives the radii for λ_k . On piece j , put

$$L_{n,j} = \sum_{v=j}^6 P_v e^{i\pi t_v n}, \quad H_{n,j} = \sum_{v=j}^6 t_v P_v e^{i\pi t_v n}.$$

For $\Phi_n(x) = e^{-i\pi n(c_j+x)}$, use $\phi_{n,0} = e^{-i\pi n c_j}$ and $\phi_{n,k+1} = (-i\pi n)\phi_{n,k}/(k+1)$. The density is then the finite series expression

$$f_j(c_j + x) = 2\pi \sum_n \Phi_n(x) \{ -\pi c_n (H_{n,j} - (c_j + x)L_{n,j}) + id_n L_{n,j} \}. \quad (\text{B.23})$$

The constant exponentials in $L_{n,j}, H_{n,j}, \phi_{n,0}$ are enclosed by (B.19). For a series $v(x)$ equal to $c_j + x$ or $z(c_j + x)$, the divided powers obey

$$V_0(x) = 1, \quad V_{\ell+1}(x) = \frac{V_\ell(x) i\pi y v(x)}{\ell + 1}. \quad (\text{B.24})$$

Finally apply the exponential recurrence to $g = i\pi m(c_j + x)$ or $g = i\pi m z(c_j + x)$, and multiply the resulting series by the density, the divided power, and, in the transformed case, λ . Truncating these recurrences after degree D gives every coefficient through D exactly at the level of formal series; omitted higher coefficients cannot enter a lower convolution coefficient.

The seven possible atoms are handled separately. At t_j , enclose their masses $e_j C P_j$ and the appropriate factor in (A.8) by the same rational operations and constant exponential bound. If their respective centers and radii are (a_j, r_j) and (g_j, s_j) , their product has radius

$$\eta_j = \langle a_j \rangle_1 s_j + \langle g_j \rangle_1 r_j + r_j s_j. \quad (\text{B.25})$$

The sum $\eta = \sum_{j=0}^6 \eta_j$ is added to the six-piece error; there is no Taylor integration tail for an atom. When $C = 0$, all these terms vanish.

All of these radii can be refined effectively. Alternating rational arctangent sums in Machin's identity enclose π to arbitrarily small width, and rational bisection of the positive square root of $3/4$ does the same for b . The kernel denominators have $|c_j + ih| \geq h = 2/5$, and $b > 4/5$. Taking real and imaginary parts of the constant exponential enclosures also encloses cosine and sine with the same radius. The nonzero sine denominators in (2.8) have modulus greater than $1/4$. Consequently sufficiently refined enclosures certify fixed positive lower bounds for the moduli of all denominators used above or in computing Q_m, D_m , and separately a positive lower bound for Q_m at pair division. For each fixed D there are finitely many operations in (B.20)–(B.24). Their radius formulas tend to zero when their input radii tend to zero, while (B.19) has radius tending to zero by refining its input and increasing its two truncation orders. Thus one may refine the rational inputs and exponential orders, recompute the displayed rational radii, and stop when any prescribed finite collection of positive coefficient and atom budgets holds. The convergence just proved ensures that this rational stopping test terminates. This is a computability statement for each specified finite set of inputs, not a claim that any of its moment sums have been evaluated.

We next make sufficient budgets explicit. Put $h_D = \lfloor D/2 \rfloor + 1$. For rational $\delta > 0$, the conditions

$$e_{j,2r} \leq \frac{\delta(2r+1)12^{2r}}{h_D} \quad (0 \leq 2r \leq D) \quad (\text{B.26})$$

on each piece and moment imply that the weighted coefficient-radius sum in (B.16) is at most δ .

We now combine these integration budgets with the pair estimate (B.6) and the Bernstein estimate from Section B.3.1.

Proposition B.5 (Two conditional Taylor budgets). *The following choices give sufficient rational error enclosures for the same finite quantities used in Appendix A.*

For a basis input at $n \in J = \{1, 3, 4\}$, a row $m \in J$, $y = 1$, and $\ell = 0, 1$, take $D = 63$, $h_D = 32$, $M = 10^6$, and $\delta = 10^{-14}$ in (B.26). There are no atoms. If Q_m, D_m have rational errors at most 10^{-25} , either entry of (A.11) has error less than $2.13 \cdot 10^{-13} < 10^{-12}$.

For the full rational finite-input domain at the start of this subsection, take $D = 255$, $h_D = 128$, $M = 10^{53}$, and $\delta = 4 \cdot 10^{-15}$ in (B.26), and require each of the seven atom-product radii in (B.25) to be at most $4 \cdot 10^{-15}/7$. Each U and X moment then has error less than 10^{-14} . For node rows $m \in I_f$, with rational Q_m, D_m errors at most 10^{-25} , either entry of (A.11) has error less than $1.8 \cdot 10^{-14}$.

Proof. The first-block majorant 10^6 is proved in Section B.5.1. Lemma B.4 gives

$$\epsilon \leq 10^6 2^{-63} + 10^{-14} < 1.19 \cdot 10^{-13}.$$

On the real interval a basis density has modulus below 214, $|\lambda| < 3$, and $\pi|z| < 10$; thus the two exact moments have modulus below 10^4 . Substitution of $H = 10^4$, $q = 7/4$, $d = 2.12$, and $\eta_Q = \eta_D = 10^{-25}$ in (B.6) gives

$$\frac{(3.12 + 10^{-25})1.19 \cdot 10^{-13} + 10^4 10^{-25}}{7/4} + \frac{3.12 \cdot 10^4 10^{-25}}{(7/4)^2} < 2.13 \cdot 10^{-13}.$$

The pointwise majorant 10^{53} for the full domain was proved in Lemma A.1. In the second case the total moment radius is at most

$$10^{53} 2^{-255} + 8 \cdot 10^{-15} < 1.728 \cdot 10^{-24} + 8 \cdot 10^{-15} < 10^{-14}.$$

For the value and first-derivative moments in (A.11), the real density is bounded by 1000, the sum of seven atom moduli by 42, and the two factors by 3 and 30. Hence both exact moments are bounded by $H = 4 \cdot 10^4$. The right side of (B.6) is then at most

$$\frac{(3.12 + 10^{-25})10^{-14} + 4 \cdot 10^4 10^{-25}}{7/4} + \frac{3.12 \cdot 4 \cdot 10^4 10^{-25}}{(7/4)^2} < 1.8 \cdot 10^{-14}.$$

All inequalities here are comparisons of the displayed rational bounds. The refinement argument above proves that their coefficient and atom budgets can be met for any prescribed finite set of these moments. $\square$

For a matrix with r rows and entry error at most ε , the induced column-norm error is at most $r\varepsilon$, and multiplication by W_η adds the factor $\|W_\eta\|_1$. In particular $r = 6$ for the J row block and $r = 12$ for the rows on $E \cap [7, 16]$. In a tabulated residual, the other summands in (A.15) are exact rationals, so its error is just the corresponding pair error. Finally, for the signed half-gaps of Lemma 4.1, the two moment errors below 10^{-14} give a power-coefficient error below $8 \cdot 10^{-14}$ and a Bernstein-coefficient error below $2.32 \cdot 10^{-12}$, as proved in Section B.3.1. These bounds specify the margins required in a Taylor-based comparison; they do not themselves assert the matrix, residual, or Bernstein values printed in Appendix A.

B.7 Direct node-Taylor representation

There is also a local Taylor representation for the direct cardinal function that does not integrate its spectral density. It applies to the full ℓ^1 class of (2.11). Write $\mathcal{N}_0 = \{0\} \cup \mathcal{N}$ and, for $m \in \mathcal{N}_0$, put

$$\rho_m = \min_{\substack{n \in \mathcal{N}_0 \\ n \neq m}} |m - n|. \quad (\text{B.27})$$

The consecutive gaps in one period of $\mathcal{N}_0$ are 1, 2, 1, 3, 2, 3. Thus the minimum exists and is positive: $\rho_m = 1$ for $m = 0$ and for residues 0, 1, 3, 4, while $\rho_m = 2$ for residues 7, 9. This is a guaranteed radius to the nearest other node; it may be smaller than the analytic radius for a particular list whose corresponding coefficients vanish.

Proposition B.6. Let $p = PR$ have a real ℓ^1 coefficient list as in (2.10). Define

$$A_k(m) = \frac{P^{(k)}(m)}{k!} = \frac{1}{k!} \sum_{v=-6}^6 P_v(i\pi t_v)^k e^{i\pi t_v m}. \quad (\text{B.28})$$

Then $A_0(m) = A_1(m) = 0$. The regular series has coefficients

$$r_j(m) = C \mathbf{1}_{\{j=0\}} + (-1)^j \sum_{\substack{n \in \mathcal{N}_0 \\ n \neq m}} \left(\frac{(j+1)c_n}{(m-n)^{j+2}} + \frac{d_n}{(m-n)^{j+1}} \right), \quad (\text{B.29})$$

and converges for $|u| < \rho_m$. For $0 < |u| < \rho_m$, the Laurent expansion is

$$R(m+u) = \frac{c_m}{u^2} + \frac{d_m}{u} + \sum_{j=0}^{\infty} r_j(m) u^j.$$

For every integer $\ell \geq 0$ the direct derivative is therefore

$$\frac{p^{(\ell)}(m)}{\ell!} = c_m A_{\ell+2}(m) + d_m A_{\ell+1}(m) + \sum_{k=2}^{\ell} A_k(m) r_{\ell-k}(m), \quad (\text{B.30})$$

where the last sum is empty for $\ell = 0, 1$.

Proof. For $n \neq m$, the geometric series for $(m-n+u)^{-1}$ and its derivative give

$$\frac{1}{m-n+u} = \sum_{j \geq 0} \frac{(-1)^j u^j}{(m-n)^{j+1}}, \quad \frac{1}{(m-n+u)^2} = \sum_{j \geq 0} \frac{(-1)^j (j+1) u^j}{(m-n)^{j+2}},$$

whenever $|u| < |m-n|$. To justify summing these series over the infinite list, fix $0 \leq r < \rho_m$ and set $q = r/\rho_m < 1$. Since $|m-n| \geq \rho_m$ for every other node, the sum of absolute values on $|u| \leq r$ is at most

$$\frac{\sum_{n \neq m} |c_n|}{\rho_m^2} \sum_{j \geq 0} (j+1) q^j + \frac{\sum_{n \neq m} |d_n|}{\rho_m} \sum_{j \geq 0} q^j = \frac{\sum_{n \neq m} |c_n|}{\rho_m^2 (1-q)^2} + \frac{\sum_{n \neq m} |d_n|}{\rho_m (1-q)} < \infty.$$

This proves absolute uniform convergence on every smaller closed disk, permits exchanging the node and Taylor sums, and gives (B.29). It also proves precisely why the nearest-other-node distance is a valid common radius.

The finite Fourier expansion (2.5) gives (B.28), and the double zero of P gives $A_0 = A_1 = 0$. Multiply its entire Taylor series by the convergent Laurent series for R . The two principal terms are canceled by the double zero; collecting the coefficient of u^ℓ gives (B.30). In particular $A_2 = Q_m$ and $A_3 = Q_m D_m$, so the cases $\ell = 0, 1$ reproduce (2.17). $\square$

For finite rational input, the constants in this direct formula can be enclosed by the rational operations above. For an arbitrary ℓ^1 input the formula is an exact convergence statement; a computable error for truncating its node sum additionally requires a bound for that list's tail. It represents the direct function p only. The transformed function K_p still has the kernel representation (2.19), and this local identity neither places K_p in the cardinal class nor replaces the exact-derivative certificates for the tabulated functions.

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